Arrange classroom desks: environment report

Environment `classroom-arrangement` (package `classroom_arrangement/`), Newton 1.5.0 + MuJoCo 3.11 (CPU), Isaac Sim 5.1 RTX path-traced images. Default robot: Franka Mobile FR3 Duo. Every number below comes from a recorded experiment run, named next to it.

0. Status in one paragraph

A furnished 71 m² classroom with 56 free rigid furniture bodies (24 Virco 785 desks, 28 Virco 9018 chairs, a tutoring table, a teacher's desk, a tote and a wastebasket; optional textbooks in the book boxes) at listed product weights, and a 216.8 kg Mobile FR3 Duo built from Franka's own description: base, Spine, two FR3 with Franka Hands, head and wrist cameras, and the TMR's four D455 cameras, two corner LIDAR scans and IMU. It is integrated with the GAR CLI, HTTP and dashboard through the existing contract (sub-step traces in GAR's trace store, the variation protocol, geometry routes, robot configuration and setup objects, an installable wheel), with a shared `base_twist` action and keyboard chords for the base and hands. "Tucked" means within ~0.12 m of the end stop, and chairs-up checks the backrest side (E20 17/17). The window wall above the sill is physical. The teacher's desk is slid 0.65 m by the robot's rear bumper within every protective limit (E15). E4's dynamic analysis is judged against the rigid-body band set by the pads (the unload lies in it or within 9 %, the rotation rate after it is 20–40 % fast and is neither a timestep nor a contact-stiffness effect). E18 states its impact model and checks it (98–99 % dissipated; a 5–8 mm rebound fails the "no rebound" expectation). The minimum command duration scales with the render load so every camera configuration stays under 200×. `reflex_stops_base` is a real option, setup objects work through `gar run`, and the stationary claims, the Spine, the reflex and each base phase are on video. The FR3 arm compensation, the Cartesian commands and the velocity reflex are measured in E25, and an arm protective stop also stops the base. **Demonstrated** (sections 7–8): rest (no body moves 0.05 mm in 30 s over 16 starts, also on video with the arms out); glide breakaway within 4.0 % of μ_s W and sliding within 2.9 % of μ_k W for 4 furniture types × 3 glides; tip-versus-slide statics in all 57 cases and in 261 of 266 sensitivity cases; base, Spine, both reflex kinds, the interlock and the base sensors to specification. The robot (scripted operator with planned collision-free arm paths, robot interface only) slides desks at μ_k W within 0–3 %, turns a desk 86°, lifts one end of a desk with three books in it and drags it 0.46 m, pushes a desk into its pod neighbour (edge gap −1.2 mm), pushes a chair in until tucked, pulls a chair out and turns it 89°, slides the teacher's desk, the tote and the wastebasket, and lifts and carries a chair (the flip onto its desk and the chairs-down lift-off are labelled phantoms). Books spill from a held, tilted desk (onset 17.7°, friction angle 16.7°); two 120 s legal-action abuse runs had no fault; every legal command in every measured camera configuration ran under 200× wall/sim with the GPU shared (worst 92×). **Not demonstrated or failing:** a two-hand push of the teacher's desk and its 90° turn, the rotation rate of a fast-tipped desk (E4), the E18 rebound expectation, five E24 cases off statics, a grip-friction control for E11, a humanoid. Of 91 evidence files, 76 pass all their checks. The 15 that do not are E4's dynamic checks (3 default runs, 4 contact-impedance variants), five E24 sensitivity cases, E18's two rebound checks and the two-hand E15 attempt. Open issues: section 12.

Videos (rendered from recorded states; contact sheets of several are in section 8)

Experiment	Shows	Assistance
E9 (nylon top, felt top, felt low)	desk slid by a hand (two hands on the legs) and the base	scripted operator
E9 control	stock franka_ros thresholds, the 20 N hand-force reflex stops the push	scripted operator
E0/E1 rest and hold	make-pod start with books: 30 s hands-off rest, then the Spine up and both arms out, 10 s hold	none, then scripted operator
E10	desk turned 86° in place by the base, both hands pinching the top (wide and top views)	scripted operator
E11	front lifted by the Spine, dragged 0.46 m on two glides, three books in this desk (low view along the lifted glides)	scripted operator
E11 weak grip	8 N per hand: the lower fingers carry the edge, the front still lifts and the desk is dragged	scripted operator
E13	chair pushed in until tucked, two hands	scripted operator
E15	62.6 kg teacher's desk slid 0.65 m by the robot's rear bumper, arms stowed	scripted operator
E15 two hands	two-hand push of the teacher's desk: the 100 N reflex and the base interlock stop it at breakaway (0.04 m)	scripted operator
E12	chair lifted 0.29 m and carried by two arms; flipped onto its desk; knock	robot, then phantom
E14	desk pushed into its pod neighbour, then the pair	scripted operator
E17	chair pulled out from its desk and turned 89° (scene laid clear of the teacher's desk)	scripted operator
E22	inverted chair lifted off its desk, turned upright, set down	phantom
E23	book tote and wastebasket slid across the floor	scripted operator
E5/E8 base	base forward/stop, turn, sideways into the wall (the stowed arm meets it, the interlock stops the base), rear-bumper wall push; a caption per phase and a view along the wall	scripted operator

Experiment	Shows	Assistance
E6 Spine, E8 reflex	Spine 0 → 0.827 m and hold; hand driven at the board: the hand-force reflex stops it, a target in the hold is refused	scripted operator
E4 (felt and nylon desk, +y, h 0.724 m)	top-edge push: felt desk tips, nylon desk slides (pushed 5 s more for the video)	phantom pusher
E2 (felt desk, soft rig)	soft-spring pull: breakaway, then sliding (this felt run does not re-stick), side view and glide close-up	phantom spring
E16	desk held tilted past its balance angle: the books slide out onto the floor	phantom
E18 (tip, drop)	felt desk pushed over (it shoves desk_14 17 mm on the way); chair dropped from 0.8 m (added drop view)	phantom

1. The activity and what was built

The action group "arrange classroom desks" (43 rows, 13 robot and 10 physics capabilities) is about relocating heavy rigid furniture against friction (pushing, dragging, lifting one end, turning, carrying) into layouts people use: rows, pods, exam spacing, chairs tucked or up on desks, clear aisles. The setting (who does it, layouts, room sizes, furniture products, glides and floors) is grounded in the sources of section 14, and every row maps to a task or an explicit exclusion (pasteboard tables, soft play equipment, floral and food arrangements are not built).

Task (`layout.TASKS`)	Real activity	What it exercises	Tier
straighten-row	desks "migrate" off their floor dots during a day	small pushes against stick-slip, 30 mm / 3° precision, tucking 6 chairs	entry
push-in-chairs	end-of-class chair tuck	non-prehensile pushing of a 5.3 kg chair under a desk	entry
seat-tutoring-group	set four chairs at a 30 × 60 in table	pulling chairs out, turning and placing them to face a table at 0.20–0.35 m	entry–intermediate
make-pod	four desks into a pod for group work	180° desk turns, ordered moves, edge-to-edge abutment ≤ 10 mm	intermediate–hard
exam-row	spread a row to 1.25 m (JCQ 11.10)	long translations holding orientation	intermediate
chairs-up	chairs inverted on desks for the cleaners	two-arm lift of a chair beyond one FR3's 3 kg rating, 180° reorientation	hard
chairs-down	the morning reverse	lifting off a desk without moving it	hard
clear-floor	fire-drill route	a 7.6 kg tote, a wastebasket, a stray chair; 0.915 m accessible route	intermediate
move-teacher-desk	free the front for a cart	144 N breakaway of the 62.6 kg desk (plus the teacher's chair it pushes) shared by two hands and the base, at the 100 N hand-force reflex	hard
group-work	three pods (extended)	long-horizon planning, 12 desks	hard (extended)

Each task has a seeded start (desk drift 5–20 cm and 2–10°, chairs out 0.4–0.8 m, loose chairs, tote and basket in the aisles), drawn conflict-free (`layout.conflicts`: no start interpenetrates, every task × 8 seeds checked by the test suite and E0), floor dots under the target glides, a short instruction and an evaluator (section 9). Variation selectors `glides` (nylon, felt, steel: Virco's 785/9018 options, which move the tip/slide boundary) and `contents` (0–3 textbooks per involved desk) change the physics; `seed` changes only the start. 10 000 catalogued variations (seeds 0–999 × 10 tasks) are split 8016/985/999 by hash (`variations.json`).

2. Room 114

Interior 7.40 × 9.60 × 3.25 m (the repository's whiteboard classroom with the back wall moved to 9.6 m; 71 m², above the Texas 700 ft² minimum). Board wall at y = 0 (2.44 × 1.22 m whiteboard, aluminium trim and tray), door at x = 2.65 with a keep-clear swing zone, three window openings in the left wall, a bookcase and cubbies at the back, VCT floor. Desks stand in three pairs of two across four rows (columns ±0.31, ±1.847, ±2.467 m; rows 2.40/3.80/5.20/6.60 m), which leaves four aisles of exactly 0.928 m: a 7.40 m room fits three pairs and four aisles 13 mm over the 0.915 m (36 in) accessible-route minimum of the 2010 ADA Standards §403.5.1 [A1]. With 0.92 m main aisles the evaluator's route check finds no route in the default layout. Students face the board; tucked chairs sit with the backrest 20 mm from the desk top's student edge (`furniture.TUCK_DY`).



Room overview

Room overview, straighten-row start (seed 0; row 3 displaced). Stills of every task's start were rendered from the room, front and plan cameras and the robot's head camera.

2.1 Furniture as rigid bodies (`furniture.py`)

Each item is one free body built from collision primitives whose density is its share of the item's mass, so mass, centre of mass and inertia follow from where the steel, plastic and wood are. **Mass basis (one rule):** the weight listed on the Ezell Company product page where Ezell lists the item (desk, chair, table), else SchoolOutlet's spec table (teacher's desk). Listed weights may include packaging; E24's −15 % case covers that.

Item	Product basis	Parts (mass)	Mass (source)	Centre of mass (body frame)
student desk	Virco 785: 24 × 18 in hard-plastic top at 29 in, 22 × 15 × 4.75 in steel book box open to the student, 1 in legs	top 3.3, book box 2.8, rails 1.6, legs 3.0, glides 0.2 kg	10.9 kg (24 lb, Ezell [V2]; other resellers 28.5–30.2 lb [V3, V4])	(0, −0.014, 0.579) m
chair	Virco 9018: 18 in seat 15 in wide, 476 × 533 × 775 mm, injection-moulded polyethylene shell, 7/8 in chrome frame under the seat	seat, back, legs, rails, back supports, glides (1.0 : 0.8 : 1.7 : 0.7 : 0.9 : 0.2)	5.26 kg (11.6 lb, Ezell [V6b]; COE Distributing 11.6 lb; SchoolOutlet 10.4 lb [V6])	(0, 0.067, 0.407) m
tutoring table	Virco 4000 (483060) 30 × 60 in, 1⅛ in particleboard top	top 23.9 (720 kg/m³), telescoping legs 5.2, aprons and anti-warp bar 3.0, glides 0.2 kg (the parts sum to the listed weight)	32.3 kg (71.3 lb, Ezell [V7b])	(0, 0, 0.656) m
teacher's desk	Virco 543 (540 Series) 48 × 30 in single pedestal, 1⅛ in top, three drawers	top 19.1 (720 kg/m³, as the table), steel pedestal 35.7, modesty panel 5.0, legs 2.4, glides 0.4 kg	62.6 kg (138 lb, SchoolOutlet, one carton [V8b])	(0.190, 0.028, 0.483) m
tote	PP tote with four 1.6 kg textbooks (one body: the books cannot shift; see section 12)		7.6 kg	(−0.021, 0, 0.130) m
wastebasket	PP, Ø 0.30 × 0.38 m (closed hull)		1.0 kg	(0, 0, 0.190) m
textbook (contents)	260 × 205 × 40 mm hardcover; lie flat in the book box, two side by side and one on top		1.6 kg	

Glides are cylinders (desk Ø 28.6 × 12 mm, chair Ø 32 × 12 mm); legs and frame tubes are capsules; tops, panels and the book box are boxes. The glide half-spacings (desk 0.275 × 0.19 m, chair 0.218 × 0.217 m) are authored (the whiteboard classroom's furniture; Virco's Revit families [V9] were downloaded but not converted) and set every tip/slide boundary, so E24 varies them by ±2 cm. The visual model is drawn from the same part list (bevelled tops, tubes, glides) except the chair's seat and back, which render as one moulded shell (the collision panels are not drawn).

3. The robot: Franka Mobile FR3 Duo (`robots/models/mobile-fr3-duo`)

Source. `franka_description` commit 7aeeddc (2026-08-17), `mobile_fr3_duo_v0_2.urdf.xacro` expanded with xacro 2.1.0 with `robot_types=['tmrv0_2','fr3','fr3']`: TMR v0.2 base (two ArgoDrive swerve units at (0.30, −0.20) and (−0.30, 0.20) m, two passive casters, the rear drive and caster on a rocker rolling ±0.16–0.18 rad), Franka Spine v0.1 (prismatic, URDF range 0–0.85 m; the simulation stops it at the datasheet's 827 mm stroke), FR3 Duo mount v0.3, Franka head v0.2, two FR3 with Franka Hands. Apache-2.0; the expanded and edited URDF and its 29 meshes are installed in the shared robot asset cache and verified against `assets.lock.json` on every load (`gar doctor` checks it).

Edits (each recorded in the package's `SOURCE.json`):

Edit	Why
mesh URIs → package-relative	self-contained package
`base_link` gets an explicit 1 g inertial	it carries the chassis collider but no `<inertial>`: Newton gave it the collider's volume × default density, +143 kg (measured: 360 kg robot)
TMR +11.46 kg battery/electronics block at z 0.12 m; Spine column +19.09 kg at its COM; Spine carriage 4.0 kg (URDF 0.01 kg)	manual R04245: TMR 130 kg, Spine 45 kg (URDF 118.5 and 21.9 kg). Total 216.8 kg vs the manual's 225 kg (−3.7 %, unreconciled remainder)
Spine 670 N, 0.3 m/s	2π·2.37 N·m / 0.020 m × 0.9 ball-screw efficiency; manual speed; the URDF's 100 N / 0.1 m/s cannot lift the 46 kg it carries
ArgoDrive 28.4 N·m / 40 rad/s; steering 30 N·m / π rad/s; Gazebo `<dynamics>` removed from TMR and Spine joints	AD-100.12 datasheet (braking torque, 2 m/s nominal, 180°/s); the controller owns drive damping
FR3 joint limits kept as in `franka_description`	they match the current Franka specification (FCI robot specifications; FR3 datasheet R02212 2.5, June 2026: A1 ±166°, A2 ±105°, A4 −176…−7°, A6 25…265°). The catalog FR3 (`robots/models/fr3`, Newton bundle) carries an older narrower set; link inertials and joint frames are byte-identical to the catalog FR3's (checked)

Catalog integration. Two models (`mobile-fr3-duo`, `mobile-fr3-single` = the same robot with the left arm removed, an authored reduced configuration), each with groups `base_steer`, `base_drive`, `base_passive`, `spine`, `<side>_arm`, `<side>_gripper`, frames (base, spine carriage, head camera, wrists, flanges, TCPs) and a `classroom` controller profile: FR3 joint stiffness from libfranka's `setDefaultBehavior` joint impedance (3000, 3000, 3000, 2500, 2500, 2000, 2000 N·m/rad; damping 100…40 N·m·s/rad authored; with the barcode profile's 600 N·m/rad the "lift" in E11 only tilts the desk 0.4°), fingers 3500/40, steering kp 400, wheels kp 300 / kd 25, Spine 1 MN/m stand-in for the ball screw, casters and rocker passive (URDF damping). Rigs `classroom-mobile-fr3-duo` and `-single`; `robots/catalog.py ENVIRONMENTS["classroom-arrangement"]` declares them and the unsupported robots with reasons (G1: the repository pins Unitree's pre-trained `unitree_rl_gym` G1 walking policy with a Newton adapter in `football/`, but it has not been tried or qualified for walking among furniture in this room or for pushing and lifting while walking, see section 12; every fixed-base rig: cannot reach across the room). A `gar.robot-setup.v1` object (or rig name) that describes one of the two qualified configurations (one instance `robot`, model `mobile-fr3-duo@1` or `mobile-fr3-single@1`, mount `floor_start`, profile `classroom`, no attachments) is accepted on every path (`gar run --robot/--config`, `registry.create`, `LaunchSpec`) through the catalogue validator `classroom_arrangement/robot_setup.py`, and anything else is refused with what it would need. The shared Newton adapter (`robots/backends/newton.py`) treats URDF `continuous` joints (wheels, steering, casters) as actuated/passive DOFs like revolute ones (no other catalog URDF has one, so no other robot's joint inventory changes).

Base sensors. The Mobile FR3 Duo carries, besides the head ZED Mini and the two wrist D405, four RealSense D455 on the TMR (front, rear, left, right), two Sick nanoScan3 safety LIDARs and an Olive Robotics IMU (manual R04245 §5.6 [D1]); the URDF has their mounting frames. Without them a policy driving a 0.72 m-wide robot through 0.93 m aisles would be blinder than the real one. `front_rgb`, `rear_rgb`, `left_rgb`, `right_rgb` are policy cameras at the URDF's `*_mounting_point` frames on `base_link` (D455 RGB, 90° HFOV, optical axes level and outward); `robot_status.base_sensors.lidar` holds the two corner scans (275° each, 0.5° steps, 0.05–10 m, ray-cast against every collider except the robot's chassis at the mounts' 0.19 m height; 1.8 ms per observation); `robot_status.base_sensors.imu` the base's angular rate and specific force at `imu_mounting_point` (10 ms low-pass; no bias or noise model). E26 checks them. The nanoScan3's safety fields are not configured on delivery ("can be set up using the Sick Safety Designer software", manual): here the scans are observations, not a protective stop.

Stowed pose. Franka publishes none for the Duo. The "stowed" pose was found by a search over the FR3 limits: both arms folded against the Spine, ≥ 20 mm from every robot part, hands ≥ 0.30 m apart, above 0.77 m, and a collision-free straight joint path to the FR3 "ready" pose. The search model left out the wrist cameras and the side wall, so candidate poses were checked in the full scene: the Duo mount angles the arms 51° outward, and a pose can put an arm into the side wall, a link into the Spine column or one link7 against the other. **The stowed robot is wider than its base:** the Duo mount puts the FR3 shoulders (link1–link3) about 0.35 m either side of the centre

line, so in a sideways approach the arms, not the 583 mm chassis, meet a wall first (E5 wall push: `base_link` stops 0.34 m from the wall); and the Duo mount, the Spine mount and the shoulders stand up to ≈ 0.6 m ahead of `base_link` at 0.35–0.85 m height, ≈ 0.2 m ahead of the TMR's front (0.40 m; measured from the built model's collision boxes), so the robot cannot push furniture with its chassis: whatever it drives into meets the arm mount first. The start-state conflict check therefore uses a 0.36 m half-width (`layout.HALF["robot"]`), and the move-teacher-desk start keeps the robot clear of the wastebasket.



Robot's head camera, make-pod start

The robot's head camera (policy view) at the make-pod start: VCT tiles, desk and chair legs, green floor dots.

3.1 Controllers (`base.py`, `world.py`)

- **Base:** the policy's `base_twist` (v_x, v_y, ω_z in the base frame, ≤ 1.0 m/s, ≤ 1.0 rad/s) is slewed at 0.8 m/s² / 1.5 rad/s²; drive i at r_i needs the contact velocity v_i = (v_x − ω y_i, v_y + ω x_i): steering angle atan2 (or ±π with the wheel reversed, whichever is nearer), wheel speed |v_i|/r × cos(steering error). Each wheel servo tracks an integrated wheel-angle reference with the wheel speed as its velocity target. **Drive torque limit:** every tick the controller decides whether the drive would accelerate the wheel in its direction of travel (or start it) or brake it, and writes the ArgoDrive limit, 16.0 N·m or 28.4 N·m (AD-100.12 [D4]), into MuJoCo's joint-level actuator force range, which clamps the total of the servo's position and velocity terms. Capping only the position lead would let the velocity term come on top, so a blocked base would push with the joint's 28.4 N·m effort limit, 1.8× the datasheet acceleration torque (E5 records 16.0 N·m). A zero command stops the reference and the servo holds the wheel, up to the motor braking torque 28.4 N·m (the drive's spring-applied holding brake, 35.5 N·m slip torque [D4], is not modelled: a parked base resists a shove ~25 % less than the real one). Odometry from the measured steering angles and wheel speeds.
- **Arms and Spine:** joint references follow targets along velocity- and acceleration-limited profiles (FR3 joints cruise at 0.90 × the datasheet speeds with 10 rad/s² from the FCI specification; Spine 0.3 m/s, 1 m/s² authored), braking to arrive, with the reference velocity as the servo's velocity target and, on the FR3 joints, an acceleration feed-forward τ_ff = M(q) a_ref from MuJoCo's mass matrix (the real controller uses its dynamic model). **Arm-link gravity is compensated:** MuJoCo's bias force on each FR3 joint (`qfrc_bias`: the gravity and Coriolis/centrifugal terms of the arm beyond that joint, from the previous 0.5 ms step) is added to the servo, as the real controller does with its dynamic model. Servo, compensation and acceleration feed-forward together are clamped to the 87 / 12 N·m joint limits (each tick the joint's actuator-force range is shifted by the feed-forward), as on the real robot whose motors supply that torque. The joint torques are internal, so the base still carries the arms; the TMR, the Spine and every payload weigh. MuJoCo's own body gravcomp is not used: it cancels each compensated body's weight through every joint above it, including the base's, so the wheels would stop carrying the arms (wheel loads 1807 N instead of 2127 N). Without compensation the servos would carry the arms' weight (15–39 N·m on joints 1–4) and a held hand would sit 14–18 mm below its command; without the feed-forward the servo overshoots its speed-limited reference when an acceleration ends, so a legal stowed → ready target would trip the velocity reflex (1.28 × the limit on joint 2). E25 measures the hold, the relative commands, large moves and wheel loads.

- **Relative commands** (`cartesian_delta`, the keyboard's hand keys) start from the arm's **commanded** pose (the FK of its joint targets, Franka's `O_T_EE_c`), not the measured one, so tracking error or a contact cannot ratchet them.
- **Franka Hand:** position mode at 50 mm/s per finger, or force mode (as in barcode_scanning): the finger reference moves at 50 mm/s, its push capped at the commanded force, ≤ 70 N per finger (GAR `inward_effort_n` ≤ 140 N total).
- **Protective stops (FR3 collision reflexes, modelled as libfranka's two kinds):** the external joint torque from the contact rows only, $\tau_{ext} = J^T f$, first-order filtered over 10 ms (a stand-in for the robot's own filtered estimates `tau_ext_hat_filtered` / `K_F_ext_hat_K`; single-tick contact impulses are not what the estimator reports), is compared (a) per joint with the profile's upper joint-torque thresholds and (b), mapped to the hand wrench w that solves $J_{tcp}^T w = \tau_{ext}$ (how the FR3 estimates the external wrench) and expressed in the hand frame, per component with the upper Cartesian force/torque thresholds; the acceleration-phase thresholds apply while the arm's reference moves. Profiles (`params.REFLEX_PROFILES`, launch option `reflex_profile`):

Profile	Joint thresholds (N·m)	Hand force / torque thresholds	Basis
`contact_x5` (default)	100, 100, 90, 90, 80, 70, 60 (above the 87/12 N·m joint limits: do not bind)	100 N, 125 N·m	franka_ros × 5: the practice of boschresearch/franka_utils `set_collision_behaviour`, whose README raises the defaults 5× "when interacting with stiff objects and expecting larger than default contact forces" [D9]
`franka_ros`	20, 20, 18, 18, 16, 14, 12	20 N, 25 N·m	franka_ros `franka_control` collision_config [D10]
`libfranka_default`	10 / 20 (nominal / acceleration)	10 / 20 N	libfranka `setDefaultBehavior` (barcode_scanning's choice)

Moving furniture needs 21–100 N at a hand (sliding a desk 21 N, the teacher's desk 72 N per hand); with the stock franka_ros thresholds the 20 N hand-force reflex stops a one-hand desk push after 69 mm (E9 control), so a furniture deployment raises the thresholds as an integrator would. Both halves, joint torque and hand force, are modelled. Velocity reflex at 1.05 × the joint limit; a stopped arm's references jump to where it is and its joint targets are refused for 1 s (as in barcode_scanning). Every reflex records its kind and the contact that caused it. **Base interlock:** a stopped arm holds its position, so a base that kept driving would push on through the stopped arms as through struts, and the hand-force reflex would count events without bounding the force (a two-hand push of the teacher's desk without the interlock peaks at 204 N with five reflexes). So any arm protective stop also stops the base (it decelerates at its limits) and base commands are refused while the arm's 1 s hold lasts (ledger `REFLEX_STOPS_BASE`, authored: an integration rule a real operator program would apply, no Franka source; backend option `reflex_stops_base=False` turns it off). With it, the 100 N threshold really bounds what two hands can push: E15's two-hand attempt stops at breakaway, and the teacher's desk is moved with the rear bumper.

- **Planning for the scripted operator** (`planner.py`, evidence only): collision-free joint paths (direct segment, else RRT-Connect + shortcuts) checked against the live scene; see section 8.
- **IK** (`kinematics.py`): each arm's chain on the moving base (carriage pose × fixed mount); a local solve from the current joints, then a continuation along the straight TCP path, then restarts from the ready, stowed and random seeds; solutions that touch the robot itself are rejected (a MuJoCo collision pass on a copy of the model in which only robot geoms collide: on the full room's ~1000 furniture geoms each check cost ~0.2 s). On the policy path a target farther from the arm's base than its sampled reach (× 1.02) is refused before any search; the continuation and restart searches stop at the first solution within 1 rad of the commanded joints or once 0.8 s have passed since the command arrived; and every command lasts ≥ 0.02 s (section 11), so planning cannot push a legal command past the 200× wall/sim limit (with every restart running, a 5 cm delta held for 0.02 s costs 259–417×, and without the reach check a 3 m target takes 4.3 s to refuse). From the stowed pose, which sits 0.06 rad from the joint 2 and joint 5 limits, most orientation-keeping Cartesian moves need a different arm configuration, which the restarts find.

4. Appearance and cameras (`render.py`, `render_worker.py`)

Isaac Sim 5.1 RTX path tracing (16 spp + OptiX denoiser for policy frames; 48–64 for stills), render-only worker that never runs a physics timeline (it refuses a stage containing a physics scene), authenticated local socket, one idle render product parked off-scene (as in barcode_scanning: products that are disabled may never return). The exported stage holds static appearance; each request carries the recorded body poses, so videos are rendered from recorded states, never re-simulated.

- Floor: 12 × 12 in vinyl composition tile (per-tile shade, fleck, recessed seams in a normal map, a wax-swirl roughness map). Walls: painted 8 × 16 in concrete block in running bond with 10 mm tooled mortar joints (generated diffuse and normal maps, eggshell roughness). Acoustic 2 × 2 ft ceiling tiles with pinholes, fissures and bevelled edges; six 2 × 4 ft troffers (rectangle lights 0.57 × 1.18 m). Outside: a pinned CC0 HDR of a suburban sports field with goals and trees (Poly Haven `suburban_football_field`, Grzegorz Wronkowski, SHA-256 `38d31700…`, downloaded and hash-checked by `render.hdri`) on the sky dome, seen through three window openings with mullions and sills (the panes are physical but not drawn: the wall above the 0.90 m sill, piers, head, frames and closed glass, is one collider and the sills are colliders, so a book placed in a drawn pier cannot fall through

it), plus a sun (intensity 1100; 1500 clips the sunlit floor patches to white). Exposure ISO 400, f/2.8, 1/100 s.

- Dressing (render-only, outside every task's working area): whiteboard with trim and tray, door, bookcase with books, a globe, cubbies with bins, bulletin board with pinned papers, printed posters (classroom expectations, periodic table, growth mindset, world map), an alphabet strip, clock, projector, coat hooks with backpacks and jackets, plants on the sills, a recycling bin.
- Furniture: "gray nebula" speckled hard-plastic tops (fleck texture with a fine normal map), CC0 oak veneer tinted maple on the tutoring table and walnut on the teacher's desk, dark steel book boxes, silver frames, navy polyethylene chair shells on chrome frames, black glides; walnut-laminate teacher's desk on grey steel with three drawers; maple tutoring table; floor dots 19 mm vinyl (green desk dots at every target glide). Textbooks render as hardcovers with cream page edges on three sides and a printed title band, the wastebasket as an open bin with a dark inside and a rolled rim (the collision bodies stay a slab and a closed cylinder).
- Robot: `franka_description` DAE visuals with their material colours; the TMR chassis is drawn two-tone (black skirt, light deck) as in Franka's product images, split at 0.26 m by clipping the triangles that cross that height (so the boundary is straight); the TMR and Spine DAEs carry millimetre coordinates without a unit tag and are scaled on export.
- Policy cameras: `head_rgb` on the Franka head bracket (ZED Mini class, 90° HFOV, frame from the URDF's `head_camera_mounting_point`), `left/right_wrist_rgb` (D405 envelope on link7, 87°) and the base's `front_rgb`, `rear_rgb`, `left_rgb`, `right_rgb` (D455, 90°, at the TMR's mounting frames), 640 × 360 by default; `head_rgb` alone is the default selection (section 11 gives the cost of the others). Diagnostic (owner only): `room`, `front`, `plan`, `chase` and `side` (both ride on the base and see only a wall when the robot stands near one, as at most task starts; the stills omit `chase`), plus fixed per-experiment views passed to the video tool. The videos are rendered at 640 × 360 per view: the GPU was shared with another environment's jobs throughout. Some fixed evidence cameras are blocked by the Spine column for part of a demonstration (E10's close view during the turn, E13's back view).

Stills of every task's start were rendered from the room, front, plan, head and left-wrist cameras.

5. Parameter ledger

Generated from `params.py` (`python -m classroom_arrangement.params`); labels: datasheet, retailer, published, asset, derived, estimate (with the range E24 varies), authored, repo. Source keys are listed in section 14.

Parameter	Value	Unit	Label	Basis
GRAVITY	9.81	m/s2	authored	standard gravity rounded as in the other environments
DT	0.0005	s	derived	sliding-glide chatter halves from 1 ms to 0.5 ms; E3 requalifies
SOLVER_ITERATIONS	50	-	repo	barcode_scanning Config
SOLVER_IMPRATIO	3	-	repo	barcode: impacts converged at 3
NOSLIP_ITERATIONS	100	-	repo	barcode: converged no-slip pass
NOSLIP_TOLERANCE	1e-10	-	repo	barcode
CONTACT_TIMECONST	0.004	s	repo	rigid contact (whiteboard/rigid_contact.py, barcode)
CONTACT_SOLIMP	(0.95, 0.99, 0.001)	-, -, m	repo	rigid contact impedance (barcode)
PAD_TIMECONST	0.005	s	repo	Franka fingertip rubber pad (label_application, anchored)
CONTACT_GAP_M	0	m	repo	boxes/planes: no early narrowphase (barcode)
MESH_CONTACT_GAP_M	0.002	m	repo	hulls/meshes: 2 mm envelope (barcode)
MAX_OBJECT_SPEED	20	m/s	authored	nothing in a classroom moves faster: a fault
MAX_OBJECT_SPIN	1000	rad/s	authored	fault guard
ROOM_X	7.4	m	authored	whiteboard classroom width; 71 m2 interior (Texas 700 ft2 minimum [S3])
ROOM_Y	9.6	m	authored	whiteboard classroom with the back wall moved 8.2 -> 9.6 m
ROOM_Z	3.25	m	authored	whiteboard classroom ceiling
DOOR_X	2.65	m	authored	door centre on the front wall (whiteboard)
DOOR_LEAF	0.93	m	authored	36 in leaf (whiteboard); swing zone kept clear

Parameter	Value	Unit	Label	Basis
AISLE_MIN	0.915	m	published	36 in clear width of an accessible route, 2010 ADA Standards 403.5.1 [A1] (literature; text not fetched)
DESK_TOP	(0.6096, 0.4572, 0.025)	m	datasheet	24 x 18 in hard plastic top [V1]
DESK_HEIGHT	0.7366	m	authored	29 in setting within the 785's 23-31 in range [V1]
DESK_MASS	10.9	kg	retailer	24 lb, Ezell listing [V2] (other resellers 28.5-30.2 lb [V3, V4]); E24 varies x0.85 / x1.15
DESK_PARTS	{'top': 3.3, 'book_box': 2.8, 'rails': 1.6, 'legs': 3.0, 'glides': 0.2}	kg	estimate	material x section, scaled to DESK_MASS
DESK_BOX	(0.559, 0.381, 0.121)	m	datasheet	book box 22 x 15 x 4.75 in [V1]
DESK_LEGS_FLOOR	(0.275, 0.19)	m	authored	glide half-spacing (whiteboard desk; no dimensioned source, the Revit family [V9] was not converted): E24 runs +-0.02 m
DESK_LEGS_TOP	(0.249, 0.17)	m	authored	slight splay of the frame (whiteboard)
DESK_LEG_R	0.0127	m	datasheet	1 in OD steel tube legs [V1]
DESK_GLIDE	(0.0143, 0.012)	m	retailer	1-1/8 in glide diameter [V3, search summary]; 12 mm tall (estimate)
CHAIR_MASS	5.26	kg	retailer	11.6 lb, Ezell listing [V6b] (COE Distributing 11.6 lb; SchoolOutlet 10.417 lb, VastMarket 11 lb item [V6]); E24 varies x0.85 / x1.15
CHAIR_PARTS	{'seat': 0.9, 'back': 0.7, 'legs': 1.5, 'rails': 0.6, 'back_supports': 0.8, 'glides': 0.2}	kg	estimate	injection-moulded polyethylene shell 1.6 kg [V5], chrome frame, scaled to CHAIR_MASS
CHAIR_SEAT_H	0.457	m	datasheet	18 in seat height [V5]
CHAIR_SEAT_HALF_W	0.1905	m	retailer	seat 15 in wide [V6]
CHAIR_OVERALL	(0.476, 0.533, 0.775)	m	datasheet	W x D x H [V5]
CHAIR_GLIDES	(0.218, 0.217)	m	authored	glide half-spacing (whiteboard chair, inside the V5 footprint; no dimensioned source): E24 runs +-0.02 m
CHAIR_GLIDE	(0.016, 0.012)	m	estimate	1-1/4 in nylon swivel glide
CHAIR_LEG_R	0.0111	m	datasheet	7/8 in chrome tube frame [V5]
TABLE_TOP	(1.524, 0.762, 0.0286)	m	datasheet	Virco 4000 Series 30 x 60 in, 1-1/8 in top [V7]
TABLE_MASS	32.3	kg	retailer	71.3 lb, Ezell listing [V7b]; parts: 720 kg/m3 particleboard top 23.9 kg, telescoping legs, aprons, anti-warp bar
TEACHER_TOP	(1.219, 0.762, 0.0286)	m	datasheet	Virco 543 48 x 30 in, 1-1/8 in top [V8, V8b]
TEACHER_HEIGHT	0.752	m	retailer	29-5/8 in, SchoolOutlet Virco 543 spec table [V8b]
TEACHER_MASS	62.6	kg	retailer	138 lb, SchoolOutlet Virco 543 spec table (1 carton) [V8b]; parts: 720 kg/m3 top 19.1 kg, steel pedestal with three drawers 35.7 kg, modesty panel, legs
TOTE	(0.52, 0.37, 0.2, 7.6)	m, kg	estimate	plastic tote with four textbooks (1.6 kg each)
BASKET	(0.15, 0.38, 1.0)	m, kg	estimate	plastic wastebasket, 0.30 m dia, 0.38 m tall
BOOK	(0.26, 0.205, 0.04, 1.6)	m, kg	estimate	hardcover textbook (desk contents variation)
GLIDE_MU	{'nylon': (0.235, 0.195), 'felt': (0.35, 0.29), 'steel': (0.12, 0.115)}	static, kinetic	published	Substratum GS-1 sled on vinyl sheet [F1] (bar chart read +-0.01); a furniture-scale pull measured mu_s 0.45 for a metal-base desk on vinyl [F2]: E24 runs nylon/steel up to 0.45
STRIBECK_V	0.002	m/s	estimate	static -> kinetic transition speed; E24 runs 1 and 5 mm/s
MU_FLOOR	0.3	-	estimate	VCT against steel tube / plastic (tipped furniture, tote, basket)
MU_PLASTIC	0.3	-	estimate	hard plastic on hard plastic (desk top-top, shell-top); x0.7-1.3
MU_STEEL	0.35	-	estimate	steel leg on steel leg; x0.7-1.3
MU_TOTE	0.35	-	estimate	polypropylene tote / basket on VCT
MU_BOOK	0.3	-	estimate	book cover on painted steel book box; spill at atan(0.30) = 16.7 deg
MU_ROBOT	0.4	-	estimate	robot covers (painted metal / plastic) on furniture

Parameter	Value	Unit	Label	Basis
MU_WHEEL	0.6	-	estimate	Vulkollan 95A tread on VCT, range 0.4-0.8
MU_PAD	0.8	-	repo	Franka fingertip pad (barcode MU pad_housing); applied to every pad contact
WHEEL_R	0.05	m	asset	ArgoDrive wheel radius (franka_description tmrv0_2, AD-100.12 datasheet)
DRIVES	{'front': (0.3, -0.2), 'rear': (-0.3, 0.2)}	m	asset	ArgoDrive positions in base_link (URDF)
WHEEL_TORQUE	(16.0, 28.4)	N m	datasheet	AD-100.12 max acceleration / motor braking torque [D4]; a stopped wheel is held by its servo up to the motor braking torque. The drive's spring-applied holding brake (slip torque 35.5 N m +-10 %, same datasheet) is not modelled: a parked base resists a shove ~25 % less than the real one
REFLEX_STOPS_BASE	True	-	authored	an arm protective stop also stops the base and refuses base commands for its 1 s hold: an integration rule (an operator program that does not stop the drive lets the base push through reflex-frozen arms); no Franka source; launch option reflex_stops_base
STEER_RATE	3.14159	rad/s	datasheet	AD-100.12 steering 180 deg/s [D4]
BASE_LIMITS	{'v': 1.0, 'w': 1.0, 'a': 0.8, 'alpha': 1.5}	m/s, rad/s, m/s2, rad/s2	authored	software limits within the manual's 1.5 m/s [D1] and the drive's 0.8 m/s2 [D4]
SPINE	{'stroke': 0.827, 'speed': 0.3, 'force': 670.2, 'accel': 1.0}	m, m/s, N, m/s2	datasheet + derived	stroke 827 mm [D1] (URDF range 0-0.85 m [D3] cut to it); 0.3 m/s [D1]; 2 pi 2.37 N m / 0.020 m x 0.9; accel 1.0 m/s2 authored (motion profile)
ARM_ACCEL	10	rad/s2	datasheet	FR3 joint acceleration limit, all joints (Franka FCI robot specifications)
ARM_SPEED_FRACTION	0.9	-	authored	joint reference cruise speed / FR3 speed limit (libfranka-style margin; with acceleration feed-forward the servo tracks within a few %; E25)
ARM_ACCEL_FEEDFORWARD	True	-	authored	FR3 joint servo adds M(q) a_ref (the real controller uses its dynamic model); without it a stowed -> ready target overshoots to 1.28 x the joint speed limit and trips the velocity reflex (E25)
ROBOT_MASS	216.7	kg	derived	URDF + reconciled TMR 130 kg / Spine 45 kg (datasheet total 225 kg [D1])
REFLEX_PROFILES	{'franka_ros': {'nominal': (20.0, 20.0, 18.0, 18.0, 16.0, 14.0, 12.0), 'acceleration': (20.0, 20.0, 18.0, 18.0, 16.0, 14.0, 12.0), 'force_nominal': (20.0, 20.0, 20.0, 25.0, 25.0, 25.0), 'force_acceleration': (20.0, 20.0, 20.0, 25.0, 25.0, 25.0)}, 'libfranka_default': {'nominal': (10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0), 'acceleration': (20.0, 20.0, 20.0, 20.0, 20.0, 20.0, 20.0), 'force_nominal': (10.0, 10.0, 10.0, 10.0, 10.0, 10.0), 'force_acceleration': (20.0, 20.0, 20.0, 20.0, 20.0, 20.0)}, 'contact_x5': {'nominal': (100.0, 100.0, 90.0, 90.0, 80.0, 70.0, 60.0), 'acceleration': (100.0, 100.0, 90.0, 90.0, 80.0, 70.0, 60.0), 'force_nominal': (100.0, 100.0, 100.0, 125.0, 125.0, 125.0), 'force_acceleration': (100.0, 100.0, 100.0, 125.0, 125.0, 125.0)}}	N m per joint; N, N m at the hand	datasheet + practice	franka_ros franka_control_node.yaml collision_config; libfranka setDefaultBehavior; boschresearch/franka_utils x5
REFLEX_FILTER_S	0.01	s	estimate	first-order filter on the contact-derived external torque and hand wrench, standing in for the FR3's own filtered estimates (tau_ext_hat_filtered, K_F_ext_hat_K); single-tick contact impulses are not what the robot's estimator reports
REFLEX_RECOVERY_S	1	s	repo	post-reflex hold before commands are accepted (barcode)
HAND	{'speed': 0.05, 'cont': 70.0, 'max_total': 140.0, 'open': 0.04}	m/s, N, N, m	datasheet	Franka Hand: 50 mm/s per finger, 70 N continuous grasp, 140 N max, 80 mm stroke [D7]
ARM_GRAVCOMP	True	-	authored	the FR3 controller compensates its own links' gravity and Coriolis terms: MuJoCo's bias force on the FR3 joints added to the servo, inside the joint torque limits; the base still carries the arms (MuJoCo body gravcomp would lift the arms off the base)
POLICY_RES	(640, 360)	px	authored	per policy camera
HEAD_HFOV	90	deg	datasheet	ZED Mini class head camera, 90 x 60 deg (D1 kit)

Parameter	Value	Unit	Label	Basis
WRIST_HFOV	87	deg	datasheet	RealSense D405 class wrist RGB (D1 kit), as barcode
BASE_CAMERA_HFOV	90	deg	datasheet	four RealSense D455 on the TMR (front, rear, left, right; manual R04245 5.6): RGB 90 x 65 deg, at the URDF's *_mounting_point frames, optical axis level and outward
LIDAR	{'fov_deg': 275.0, 'step_deg': 0.5, 'range_m': (0.05, 10.0), 'mounts': ((0.3275, 0.2175, 0.19065, 45.0), (-0.3275, -0.2175, 0.19065, 225.0))}	deg, m	datasheet / authored	two Sick nanoScan3 (manual R04245 5.6) at the URDF's lidar_*_mounting_point frames: 275 deg scanning angle (datasheet); 0.5 deg steps (authored, the scanner resolves 0.17 deg); ranges capped at 10 m (the room; the scanner measures to 40 m); scan centre pointing diagonally outward from each corner (authored from the frames' placement); ray-cast against the collision geometry, the robot's chassis excluded; the safety fields are not configured on delivery (manual), so the scans are observations only
IMU_FILTER_S	0.01	s	authored	first-order low-pass of the base IMU's specific force (Olive Robotics IMU on the TMR, manual R04245 5.6; at imu_mounting_point)
EVAL	{'window_s': 1.0, 'glide_tol': 0.03, 'yaw_tol_deg': 3.0, 'pod_gap': 0.01, 'pod_lateral': 0.03, 'tuck_depth': 0.3, 'tuck_lateral': 0.08, 'tuck_yaw_deg': 10.0, 'use_range': (0.35, 0.55), 'ready_range': (0.2, 0.35), 'facing_deg': 15.0, 'exam_pitch': 1.25, 'upright_deg': 3.0, 'rest_motion': 0.002, 'inverted_deg': 175.0, 'inside_edge': 0.05, 'glide_load_frac': 0.05, 'up_side_deg': 30.0}	m, deg	authored	floor dots hidden under a glide up to 24 mm; JCQ 11.10 exam spacing 1.25 m (a UK rule; no U.S. seat pitch found); tuck_depth 0.30 m: the seat front at least 0.30 m under the top, i.e. within ~0.12 m of the fully tucked end stop (0.418 m, backrest 20 mm from the edge) so the walkway stays clear (0.10 m would pass a chair 0.30 m out); up_side_deg: an inverted chair's backrest within 30 deg of the student side

Parameter	Value	Unit	Label	Basis
EVAL_GROUP_WORK	{'seat_front': (-0.15, 0.35), 'as_found': (0.1, 10.0), 'as_found_furniture': (0.06, 5.0), 'as_found_full': (0.3, 45.0), 'loose_as_found': (0.2, 10.0), 'loose_full': 0.5, 'pod_lateral': 0.06, 'pod_widest': 0.0419, 'toppled_deg': 30.0, 'turn_share': 0.3, 'turn_zero_deg': 90.0, 'approach_share': 0.3, 'approach_radius': 2.5, 'chair_reach': 0.2665, 'route_floor': 0.815, 'weights': {'desks': 0.6, 'abutment': 0.15, 'chairs': 0.25}, 'fail_scale': 0.9, 'place_charge': 0.05, 'penalty': {'route': 0.5, 'safety': 0.5}}	m, deg, -	authored + derived	group-work: seat_front: a chair at a pod desk is pulled out with its seat front from 0.15 m under the top edge (half of tuck_depth, so not tucked; make-pod's start chairs, 'pulled out by the students (in use)', are 0.12 m under) to 0.35 m beyond it (seat-tutoring-group's ready-to-sit limit); as_found: each chair place the pods do not use still holds a chair within 0.10 m and 10 deg of where one stood, upright (inside the catalogue's own 'tucked'); as_found_furniture: each desk, teacher's desk and table place within 0.06 m and 5 deg (push-in-chairs' 'do not move the desks' is 0.05 m / 5 deg, plus 1 cm; at 0.10 m / 10 deg 36.7 % of straighten-row's 'drifted off their places' desks counted as 'as it is', at 0.06 m / 5 deg 3.9 %; the route check only catches a route closed everywhere, not a narrowed aisle); the tote and wastebasket within 0.20 m (clear-floor's 'in its place') and tilted <= 10 deg; as_found_full / loose_full (authored): the displacement at which a place's charge is complete; pod_lateral (derived, 2 x glide_tol): facing desks may stand as far apart sideways as their two dot tolerances allow; pod_widest (derived, pod_gap + desk width x sin yaw_tol): the closest-gap limit plus the opening of one desk at the slot's full 3 deg against a square partner (a pod touching at one corner with that twist opens 31.9 mm; at the 10 mm closest gap the same twist opens 41.9 mm); reward: progress = weighted mean credit of the 12 desk places (board-side places: turn_share for facing their way, from 0 at turn_zero_deg off to 1 within yaw_tol, wherever the desk stands; of the rest, approach_share for a desk so turned coming within approach_radius of the place, centre distance, which covers the front band where desks are turned), 6 front-edge meetings and 12 chairs (fall-off over chair_reach, half the chair's depth); score 1 on success, else fail_scale x max(0, progress - place_charge x the charges of the rest of the room) x penalties: place_charge is one desk place's share of progress (0.60/12), so putting a displaced item back is worth what placing a desk is, at any stage; route x1 at 0.915 m falling to x0.5 at route_floor (ADA 2010 403.5.1 exception: 32 in / 815 mm, the narrowest any accessible route may be), x0.5 for anything toppled or in the door swing

6. Contact, friction and numerics (`world.py`)

- Newton 1.5 `SolverMuJoCo` on MuJoCo's CPU backend, float64 state kept between steps (update interval 0), implicitfast, elliptic cones, Newton solver 50 iterations, impratio 3, no-slip 100 iterations at 1e-10 (as in barcode_scanning), **dt = 0.5 ms** (E3 compares 1.0, 0.5 and 0.25 ms).
- **Sliding force: what fluctuates.** An undamped stiff (5 kN/m) pull rig swings the measured force by 7–78 % RMS: the spring and the item ring at √(k/m)/2π (1.4–4.9 Hz, the measured spectral peak), and constant sliding friction removes no energy from that oscillation. With the stiff rig critically damped (`scripted.SpringPull(zeta=1)`), the mean sliding force is unchanged (within 4 % of μ_k W) and a residual RMS remains (E2 table). The spring force recorded every 4 ticks is a 500 Hz record whose Nyquist limit is 250 Hz, too coarse to identify the residual's frequency. **What it is, read every tick:** reading the glide contacts every 0.5 ms tick instead (E2 `glide_contact_while_sliding`, E4 `trailing_ticks_unloaded_while_sliding`): a sliding glide's contact is a train of impulses, not a continuous load. In slow steady sliding (the stiff rig, 50 mm/s) each glide carries no load in 20–61 % of the ticks over the 12 item/glide combinations (the nylon desk: 34 % and 47 % for its two glide pairs), in gaps of a median 1–7 ticks (at most 15), and moves 6–38 μm vertically, while the glides' mean total load stays at 1.000 W (0.9999–1.0003); at the desk speeds of the E4 force ramp (up to about 1 m/s) the trailing glides are unloaded in 85–96 % of ticks (hops of 0.45–0.9 mm about every 6 ms). Means (friction, breakaway, the tip/slide statics) are right; the instantaneous contact forces a force-sensing policy or a sparsely sampling probe would see are impulsive. The reflexes' 10 ms filter averages 3–20 of these impulses, standing in for the FR3's own filtered estimates. The slow-sliding force RMS halves with each halving of dt (E3), which identifies that fluctuation as numerical; the intermittency of fast sliding does not go away with dt (84–97 % unloaded ticks at 0.5, 0.25 and 0.125 ms) or with the contact time constant (2–8 ms, E4), and neither does the faster tip rotation that goes with it (section 7, E4).

- Rigid contact: solref (0.004 s, 1), solimp (0.95, 0.99, 1 mm) on every geom except the Franka fingertip pads (the repository's anchored 5 ms rubber, scaled by finger mass); hard joint stops on the FR3 joints, fingers, Spine and rocker; contact envelopes 0 for boxes/planes, 2 mm for hulls (as in barcode_scanning).
- **Pair-specific friction.** Glides, drive wheels, fingertip pads and textbooks have MuJoCo geom priority 1, so their own coefficient governs each of their pairs (MuJoCo's default takes the maximum of the two geoms: 0.30 glides on a 0.7 floor would slide at 0.7).
- **Static → sliding.** Every glide's coefficient is rescheduled every tick from its slip speed over the last tick: $\mu(v) = \mu_k + (\mu_s - \mu_k)\cdot\exp(-(v/v_c)^2)$, $v_c = 2$ mm/s. E2 measures the result.
- **MuJoCo midphase disabled.** The per-body BVH midphase silently drops the rear glide–floor contacts of every chair after the first step (the chair's inertial frame is strongly rotated); with it on, all 29 chairs sink through the floor at the back and fall onto their backrests (48°) during the reset settle. A standalone chair reproduces it (stands with the midphase off, falls with it on); barcode_scanning found the same defect. Cost: physics 3.9× → 5.9× real time in the furnished room.
- **Single-contact CCD pairs.** MuJoCo warns at startup that capsule–cylinder, capsule–mesh, cylinder–cylinder, cylinder–box and cylinder–mesh pairs get at most one contact under multi-CCD. Checked against what touches what: every glide–floor and wheel–floor pair is cylinder–plane (analytic, up to 4 contacts) and every furniture resting contact used by a task is box–box, box–plane or capsule–plane; the single-contact pairs arise only transiently (a leg brushing another desk's glide, a robot link against a glide).
- Faults stop the world: non-finite state, any furniture body faster than 20 m/s or 1000 rad/s, or a MuJoCo warning (bad qacc/qvel/qpos, constraint or contact buffer full), checked every 20 ticks.
- The reset settles until every free body and the robot base have stayed under 2 mm/s for a whole 0.1 s window (at least 1 s, at most 5 s; all starts settle in 1.0 s).

7. Physical evidence (furniture, robot mechanisms)

Assistance labels: **none** = nothing but gravity and the world; **phantom** = a labelled external wrench on a body (`scripted.Phantom`, `SpringPull`, E4 `Pusher` and `ForceRamp`, `PoseServo`) or a privileged placement; **scripted operator** = the robot driven through its own interface (base twists, IK joint targets every 20 ms, Spine targets, Franka Hand commands) by a script that plans from ground-truth furniture poses. None of these is reachable from the policy API. Every video is rendered from the run whose numbers are quoted. 76 of 91 evidence files pass all their checks; the 15 that fail are E4's dynamic checks (7 files), five E24 variants, E18's two rebound checks and the two-hand E15 attempt (all discussed below). The robot demonstrations E9, E12–E14, E17, E19, E22, E23 and E25 were run before the window-wall and sill colliders, the 0.752 m teacher's desk and the base sensors were added. Those changes do not touch the bodies, places or forces of these runs: none reaches the window wall or involves the teacher's desk, and the sensors add no forces.

E0/E1: start states and rest (assistance: none)

Every task's start (seed 0), plus straighten-row with felt and steel glides, straighten-row seed 3, push-in-chairs seed 5 and make-pod with three textbooks lying flat in each of four desks (15 runs, plus the rest-and-hold video run; 56–68 free bodies + robot): settled in 1.0 s (residual < 2 mm/s), deepest interpenetration 0.23 mm (a caster on the floor), every piece of furniture upright (worst tilt 0.001°), all four wheels and casters loaded (front caster 616 N, front drive 803 N, rear caster 290 N, rear drive 418 N: 2127 N = 216.8 kg × g), no protective stop, and over 30 s hands-off **no body moved by as much as 0.05 mm** (the JSONs round to 0.1 µm: 0.0000 mm). The books lie flat in the box (`scene.py`): spawned side by side three to a box that holds two, they overlap, lean (0°, 11°, 20°) and creep 0.07 µm/s.

Rest on video. The rest-and-hold video records the make-pod start with three books in each of four desks: 30 s hands-off (no body moves 0.1 µm), then the Spine rises 0.5 m and both arms move out to E25's forward pose and are held 10 s: furniture 0.0 mm, hands within 0.039 mm of their commanded pose, no protective stop. An independent 60 s rest, 30 s arms-out hold and 20 s chair-in-the-air pinch on a start no evidence file uses agree (nothing above 0.07 mm; the pinch creeps ~3 µm/s).



E1 rest

E1: largest net motion of any body in 30 s for each start (log scale; red: the 0.5 mm acceptance).

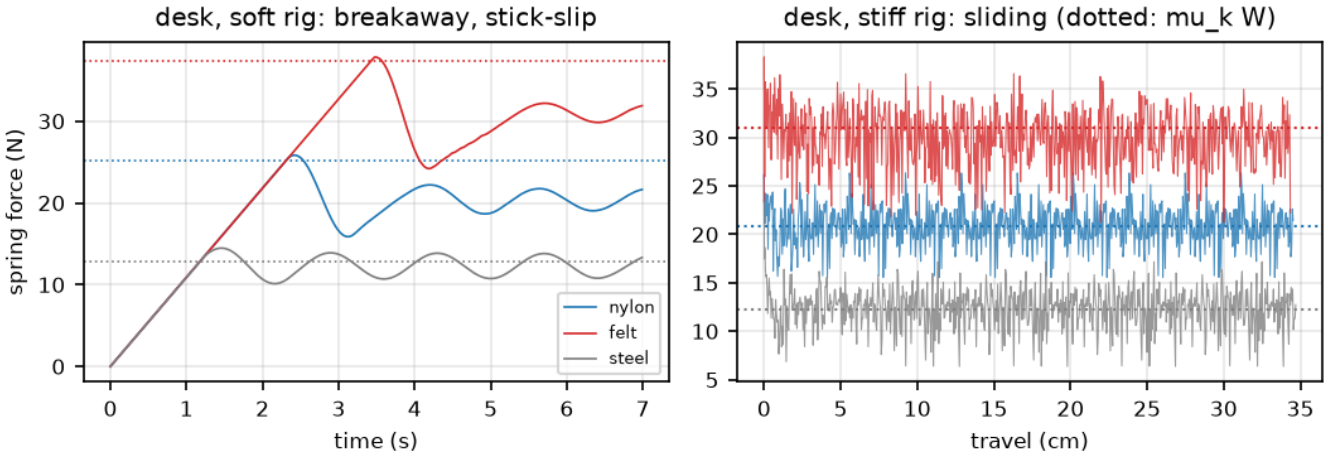
E2: glide friction (assistance: phantom spring)

Each item stands on its four glides and is pulled at 0.10 m height through a spring whose free end moves at 50 mm/s. **Soft rig** (k = 20 N/m per kg, undamped): the force ramps until the item breaks away (peak = static friction, µ_s W) and then stick-slips with drops of about 2(µ_s − µ_k)W. **Stiff rig** (5 kN/m, critically damped): near-constant-speed sliding, mean = µ_k W.

Item / glides	breakaway / µ_s W	sliding / µ_k W (stiff)	sliding RMS (stiff)	first stick-slip drop / 2(µ_s − µ_k)W
desk / nylon	1.012	0.998	12.5 %	8.2 / 8.6 N
desk / felt	1.007	0.971	10.1 %	11.4 / 12.8 N
desk / steel	1.034	1.000	17.2 %	3.8 / 1.1 N
chair / nylon	1.014	0.995	9.9 %	4.2 / 4.1 N
chair / felt	1.009	1.001	11.4 %	5.9 / 6.2 N
chair / steel	1.040	0.997	15.4 %	1.9 / 0.5 N
table / nylon	1.013	1.000	7.4 %	25.5 / 25.4 N
table / felt	1.008	1.000	10.3 %	34.1 / 38.0 N
table / steel	1.036	1.014	6.0 %	11.2 / 3.2 N

Item / glides	breakaway / μ_s W	sliding / μ_k W (stiff)	sliding RMS (stiff)	first stick-slip drop / 2(μ_s − μ_k)W
teacher's desk / nylon	1.013	0.988	1.7 %	51.3 / 49.1 N
teacher's desk / felt	1.008	0.983	1.4 %	74.0 / 73.7 N
teacher's desk / steel	1.039	0.993	2.1 %	24.1 / 6.1 N

Breakaway within 0.7–4.0 % of μ_s W (read where the item first moves faster than the 2 mm/s Stribeck velocity) and sliding within 2.9 % of μ_k W for all 12 combinations. The stick-slip drops match 2(μ_s − μ_k)W for nylon and felt (within 11 %); for steel glides, whose μ_s exceeds μ_k by only 4 %, the drops are 3–4× larger than that step: the sawtooth there is set by the spring and contact dynamics, not by the friction step. **Stiff-rig fluctuation:** an undamped rig rings at √(k/m)/2π and swings 7–78 % RMS; with critical damping the RMS is 1.5–17 % around the right mean. The spring force is recorded at 500 Hz, too coarse to identify a frequency; read every tick, the glide contacts are intermittent (section 6).



E2 friction

E2, desk: soft rig (left: breakaway, then stick-slip; dotted: μ_s W) and critically damped stiff rig (right; dotted: μ_k W).

E3: timestep convergence (assistance: phantom rigs)

dt	rest: largest net motion	desk sliding / μ_k W (stiff rig)	sliding RMS	desk breakaway / μ_s W	top-edge push, nylon / felt
1.0 ms	0.0 mm	1.000	20.6 %	1.019	slides / tips
0.5 ms (default)	0.0 mm	0.998	12.5 %	1.019	slides / tips
0.25 ms	0.0 mm	0.976	7.7 %	1.020	slides / tips

The physical outcomes do not depend on the step; the residual high-frequency sliding fluctuation roughly halves with each halving of dt, which is what identifies it as numerical (contact-level) rather than physical.

E4: tip versus slide (assistance: phantom pusher / force)

Statics. A frictionless unilateral spring-damper pusher whose end moves at 50 mm/s acts at a body-fixed point of a desk (five heights × three directions) or a chair (top rail, 0.60 m, seat, pushed forward at the top rail). Statics: the item tips before it slides iff μ_s > d/h, with d the horizontal distance from the centre of mass to the outer edge of the leading glide pads and h the push height. **All 57 cases (19 per glide type) on the predicted side**; closest margins 3.7–4.0 % (felt desk at 0.60 m, felt chair at the seat), and no case passes only because of the ±6 % tolerance band the probe allows.

Dynamics, with a controlled input. Predicting tipping from the desk's own measured peak acceleration would be circular, because that acceleration already contains the tip. So a phantom applies a **horizontal force** at height h above the centre of mass (frictionless point load), ramped at 50 N/s until the desk breaks away and then at 200 N/s, in the cases where statics says it slides first and the predicted onset is reachable before the desk runs into the next row (F* < 160 N). While it slides, m a = F − μ_k W, and the moment balance about the leading glide line gives the trailing glides' load as a straight line in F, N_t(F) = [W (d − μ_k h_c) − F (h − h_c)]/L, which reaches zero at **F* = W (d − μ_k h_c)/(h − h_c)**; past F* the excess moment turns the desk, and a rigid desk with the model's inertia about the pivot (I_p = 4.86 kg·m²) would reach 1° at F_1°,pred = F* + ṙ t₁ with t₁ = (6 θ I_p / (ṙ (h − h_c)))^⅓.

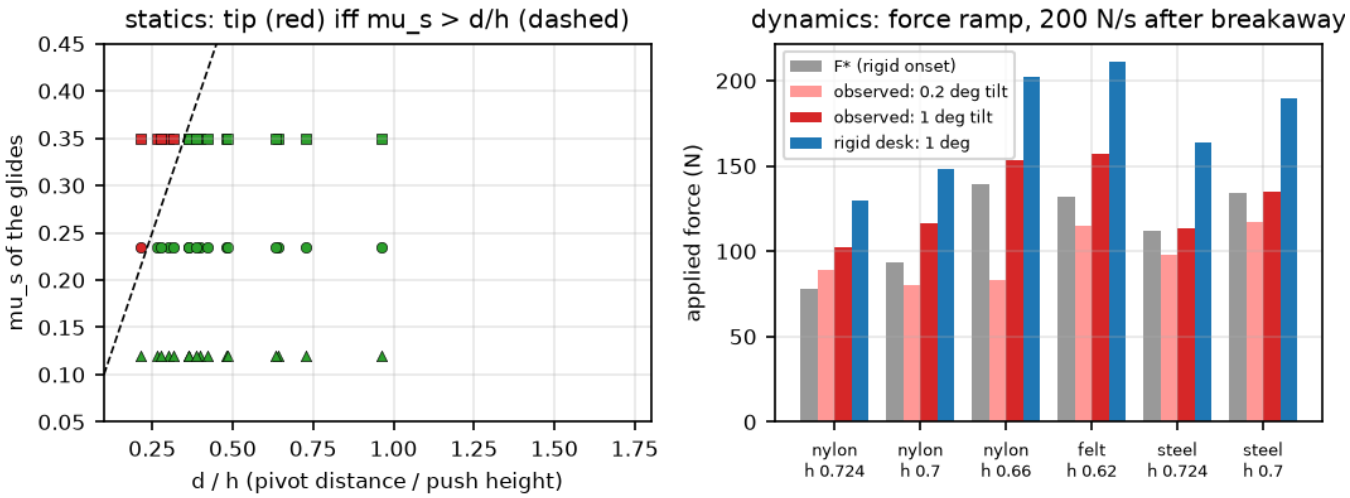
Case	breakaway / μ_s W	F*′ – F* (rigid band)	trailing load < 2 % W at	tilt 0.2° at	1° tilt at (rigid estimate)	trailing glides unloaded
nylon, h 0.724 m	26.12 / 25.13 N	67.1 – 77.7 N	61.4 N (0.79 F*) (below the band)	89.1 N (1.15 F*)	102.6 N (129.7 N)	91 %

Case	breakaway / μ_s W	F*′ – F* (rigid band)	trailing load < 2 % W at	tilt 0.2° at	1° tilt at (rigid estimate)	trailing glides unloaded
nylon, h 0.7 m	26.12 / 25.13 N	80.5 – 93.2 N	74.9 N (0.80 F*) (below the band)	80.1 N (0.86 F*)	116.6 N (148.4 N)	90 %
nylon, h 0.66 m	26.12 / 25.13 N	120.4 – 139.3 N	—	83.1 N (0.60 F*)	153.1 N (202.5 N)	92 %
felt, h 0.62 m	37.25 / 37.43 N	94.1 – 131.7 N	101.5 N (0.77 F*)	115.2 N (0.88 F*)	157.2 N (211.1 N)	96 %
steel, h 0.724 m	14.00 / 12.83 N	101.4 – 112.0 N	98.7 N (0.88 F*)	98.0 N (0.88 F*)	113.5 N (164.0 N)	84 %
steel, h 0.7 m	14.00 / 12.83 N	121.5 – 134.2 N	117.2 N (0.87 F*)	117.0 N (0.87 F*)	135.0 N (189.5 N)	86 %

Reading. A rigid desk sliding flat on four 28.6 mm pads does not fix where its leading pads carry their load, so its trailing load can reach zero anywhere in the band [F*′, F*]: F*′ with the load at the pad centres (d − r_g, r_g = 14.3 mm), F* at the rims. Rotation can only start once the trailing load is gone. At the default 0.5 ms step the felt and steel cases unload inside the band and nothing rotates before it; the nylon cases unload 7–9 % below F*′ (0.79–0.80 F* against 0.864), and the steepest nylon case (h 0.66 m) tilts 0.2° already at 0.60 F*. The nylon h 0.724 case re-run at 0.25 and 0.125 ms unloads at 0.89 and 0.87 F*, inside the band, so that part is a timestep effect of the default dt. What does not converge is the **rotation rate**: once the desk rotates it reaches 1° at 0.69–0.79 of a rigid body's cube-root estimate at every step (0.5 ms here; 0.79–0.81 at 0.25 and 0.125 ms) and at every contact time constant (0.61–0.80 at 2 and 8 ms against the default 4 ms, below), with the trailing glides off the floor in 84–97 % of the ticks throughout:

Case	contact time constant	trailing load < 2 % W	1° tilt / rigid estimate	unloaded
nylon, h 0.724 m	2 ms	0.84 F*	0.80	93 %
nylon, h 0.7 m	2 ms	0.95 F*	0.65	92 %
nylon, h 0.66 m	2 ms	0.87 F*	0.65	94 %
nylon, h 0.724 m	8 ms	0.81 F*	0.80	87 %
nylon, h 0.7 m	8 ms	0.77 F*	0.79	88 %
nylon, h 0.66 m	8 ms	0.84 F*	0.73	88 %
felt, h 0.62 m	2 ms	0.66 F*	0.61	97 %
felt, h 0.62 m	8 ms	0.95 F*	0.63	96 %

So the faster rotation is neither the step nor the contact stiffness; whether the cube-root estimate (rotation about a fixed leading-pad rim with the full ramp acting from F*) is too crude or the hopping contact rotates the desk faster was not established. The probe's checks are: unloading inside the rigid band (passes for felt and steel, fails for nylon at 0.5 ms), no rotation before the band (fails for nylon h 0.66 m), the 1° tilt within 15 % of the rigid estimate (fails everywhere: the open item) and breakaway within 5 % (steel reads 9 % high in this fast test, 3.4 % in E2). Checks against the rim line alone cannot tell a defect from the indeterminacy, so the band is used. What the evidence supports: statics exactly (57/57; slow pushes, E9), dynamic unloading inside the rigid band within 7–9 % (nylon, default dt) or exactly (felt, steel; nylon at finer dt); not supported: the rotation rate after the onset.



E4 map

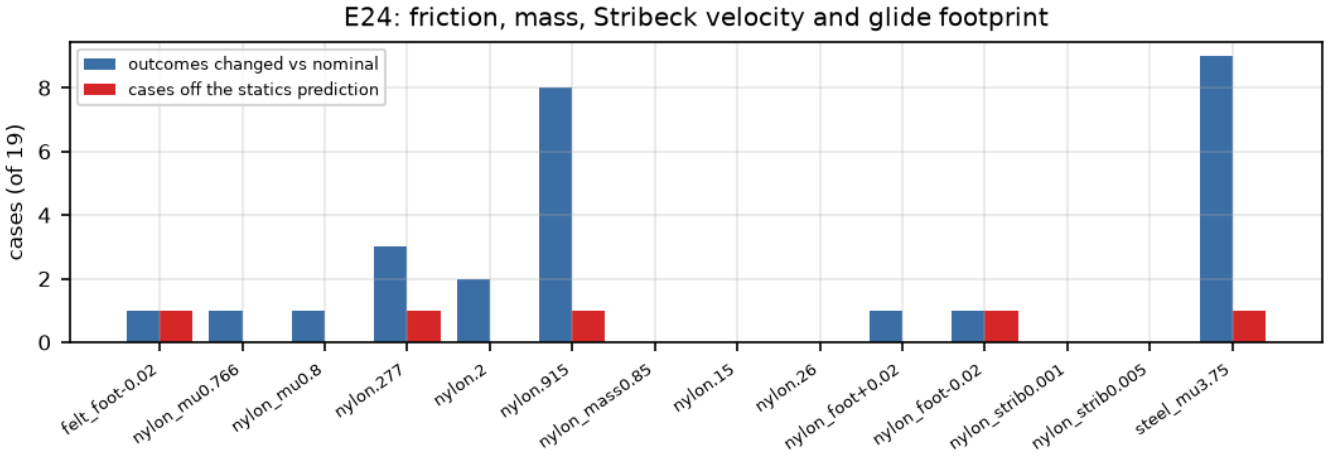
E4. Left: the 57 static cases (red: tipped), on the predicted side of μ_s = d/h (dashed). Right: dynamic tip onset under a controlled force ramp against the predicted F.*

E24: sensitivity to the least certain inputs (assistance: phantom)

A friction change of ×0.8/×1.2 does not span the published disagreement: the glide coefficients come from a small sled test [F1] (nylon μ_s 0.235), while a furniture-scale pull measured μ_s 0.45 for a metal-base desk on vinyl [F2]; and the glide footprint, which sets d, is authored. Every E4 static case was re-run with each input changed alone:

Variant	statics followed	outcomes that changed from nominal
nylon μ_s 0.18 (×0.766)	19/19	chair pulled at the top rail now slides
nylon ×0.8	19/19	chair pulled at the top rail now slides
nylon ×1.2	19/19	desk −y at the top edge, chair pulled at 0.60 m now tip
nylon μ_s 0.30 (×1.277)	18/19	+ desk +y at the top edge (the miss: margin −0.4 %, inside the band)
nylon μ_s 0.45 (×1.915, furniture-scale [F2])	18/19	**8 of 19: the desk tips when pushed at its top edge (all three directions) and at 0.60 m; the chair tips when pulled at 0.60 m or at the seat or pushed at the top rail**
steel μ_s 0.45 (×3.75)	18/19	9 of 19, the same as nylon at 0.45 plus the chair top-rail pull
mass ×0.85 / ×1.15 / ×1.26 (×1.18 and ×1.26 are the other two listings of the desk, 28.5 and 30.2 lb [V3, V4])	19/19 each	none (breakaway scales with W)
Stribeck velocity 1 / 5 mm/s	19/19 each	none
glide half-spacing +2 cm	19/19	chair pulled at the top rail now slides
glide half-spacing −2 cm	18/19	desk −y at the top edge now tips (the miss: margin −0.2 %)
felt, glide half-spacing −2 cm	18/19	desk +y at 0.60 m now tips

The model follows statics in 261 of the 266 changed cases (14 variants × 19); two misses are within 0.4 % of the boundary, and three (nylon and steel at 0.45, felt with the narrower footprint) are 6–9 % on the tipping side but slid: with the larger μ_s − μ_k gap of those variants the Stribeck schedule lets the pivot glides creep and drop toward μ_k before the item rotates. **What this means for the tasks:** with the sled coefficients a desk can be pushed at its top edge (it slides); if real glides behave like the furniture-scale pull (μ_s 0.45), a top-edge push tips a desk and the push must go at or below about 0.45 m (the legs, as E9's `felt_low` case does). The default stays the sled value because [F1] measured the three glide types this environment offers on the same surface, while [F2] is one metal-base office desk; the `glides` selector and this table expose the difference rather than hide it. Mass changes no outcome (both sides of μ_s W h ≷ W d scale with W), but every push force and reflex margin scales with it: at the heaviest desk listing (13.7 kg, ×1.26) E9's one-hand slide needs 26 N instead of 21 N and E11's front-edge lift 68 N instead of 54 N, both still under the default 100 N hand-force threshold. Ezell's 24 lb is kept as the one-rule basis (Ezell lists every item but the teacher's desk); its 15 lb for the laminate-top sibling 785MBB is implausible for the same frame and book box, which is a reason to doubt Ezell's weights as a class, so ×1.26 is covered here rather than adopted. The published spread of glide friction is about 2×, not ±20 %.



E24

E24: outcomes changed from nominal (blue) and cases off the statics prediction (red) per variant.

E5/E6/E8: base, Spine, reflex (assistance: scripted operator)

- **Base:** 0.8 m/s forward: moved 1.500 m, stopped in 0.389 m from the zero command ($0.8^2/(2 \times 0.8) = 0.40$ m), peak speed 0.815 m/s; 180° turn at 0.8 rad/s: −173.4° (peak 0.86 rad/s); 0.5 m sideways: 0.477 m. **Sideways into the front wall** at 0.3 m/s: the stowed **left arm meets the wall first** (link2/link3 on the wall and the whiteboard's marker tray) while `base_link` is still 0.45 m from it; the arm's 100 N hand-force reflex fires once and the **base interlock** stops the base (speed 0 at the end of the command). Without the interlock the base pushes on: 8 reflexes, the chassis leaning 5.7° propped by that arm. **Backwards into the same wall** at 0.3 m/s for 2.5 s after contact: the chassis's rear bumper touches (the arms stand 0.16 m inside it); the drive wheels push at their **16.0 N·m** driving limit, the base stops (1.9 mm/s), the slipping rear drive is braked at up to 16.6 N·m (inside the 28.4 N·m braking limit), no reflex, no lean (0.03°). Odometry is off by 3.8 cm after the free moves and

by 0.47 m after the wheels have slipped against the wall. The video captions each phase and adds a view along the wall that shows both contacts. Whether a real TMR's nanoScan3 would stop the base before such contacts depends on safety fields the integrator configures (they are not set on delivery); here the scans are observations (E26).

- **Spine:** 0 → 0.827 m (the datasheet stroke) at 0.3 m/s with the arms stowed: tracking error 0.31 mm, peak speed 0.302 m/s; holding 5 s at the top: no measurable creep.
- **Reflex:** the right hand driven toward a target 0.10 m behind the whiteboard: with the default `contact_x5` profile the **Cartesian** reflex stops the arm when the hand-force estimate reaches 100.5 N against its 100 N threshold (fingertip/link7 on the board face, cause recorded); the next joint target during the 1 s hold was refused and one after it accepted. (E9's control shows the stock franka_ros 20 N threshold; E15 shows the 100 N threshold, with the base interlock, stopping a two-hand push of the teacher's desk.)

E25: arm control (robot interface only)

Checks of the arm gravity compensation, relative commands and the velocity reflex on a legal target. Robot parked in the open front teaching area, nothing within reach, no phantom.

Check	Result
hold, left hand at two forward poses for 2 s	servo 0.0–0.2 N·m; the compensation carries 7–29 N·m (= MuJoCo's gravity bias to 0.01 N·m); hand within 0.04 mm of its command; wheel loads = robot weight (1.000)
6 × `cartesian_delta (0,0,0)`, 6 × `(0, +0.02, 0)`, 6 × keyboard S (0.5 s), public backend	hand height unchanged (0.0 mm net, 0.1 mm largest excursion); the six +2 cm deltas moved it 0.120 m
single `named_joint_target` stowed → ready, ready → stowed, stowed → forward reach, back (4 s each, each arm)	no reflex; peak joint speed 0.91–0.92 × the limit

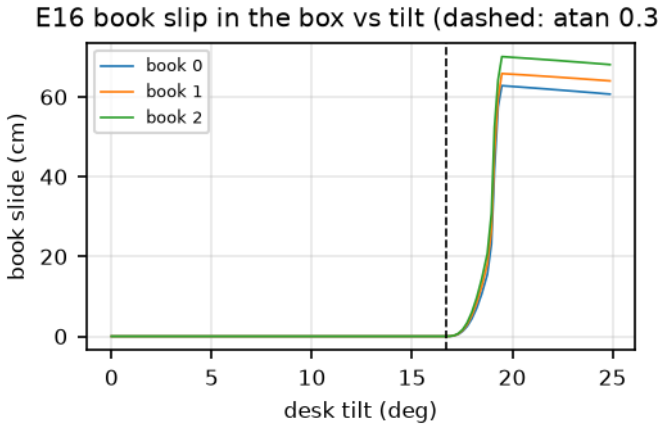
These rest on three controller parts (section 3.1): compensation of the arm's gravity and velocity terms in the joint servos, an acceleration feed-forward with a 0.90 cruise fraction, and relative commands from the commanded pose. Without them the servos carry 15–39 N·m on joints 1–4 and the hand holds 13.6 / 17.6 mm below its command, the 18 relative commands above lower the hand 0.290 m (16 mm per command), and the large moves trip the velocity reflex on joint 2 (1.05 × the limit; peak 1.28–1.30 × after the stop).

E26: base sensors (robot interface only)

At E5's start each corner LIDAR's beam toward the front wall reads the mount's perpendicular distance to it to 0.5 mm (1.368 / 1.3676 m and 0.933 / 0.9325 m); beams toward the window wall stop at the nearer furniture in between, and no beam ends inside the chassis (closest returns 0.835 and 0.135 m). The IMU reads (0, 0, 9.81) m/s² and zero rate at rest, +0.80 m/s² while the base accelerates at its 0.8 m/s² slew, and 0.847 rad/s while it turns at a commanded 0.8 rad/s (the base's own yaw-rate overshoot). The four D455 views of the make-pod start were rendered as stills.

E16: desk contents spill (assistance: phantom)

desk_13 with three textbooks lying flat in its book box is tilted toward the box's open side by a phantom force normal to its top at the front edge, servoed on the tilt: 1°/s up to 28°, held there (past the desk's 20.6° balance angle about its leading glides, so the phantom holds it back), then lowered again. **Onset of slip** (a book moved 30 mm toward the opening in the desk's frame): 17.7°, 17.8° and 17.8° (friction angle atan 0.30 = 16.7°; acceptance ±3°). **Spill:** all three books slide out and lie on the floor 18.8 s after the start (the tilt was then ~18.8°: they leave about 1 s after the onset, before the desk reaches its balance angle); the desk is lowered back onto its glides (final tilt 0.0°). The phantom is held and lowered so the desk does not fall over with the books still in the box. It pushes along the desk's own normal, which keeps a constant lever: a vertical push loses its lever near 30° of tilt, where the edge passes over the pivot.

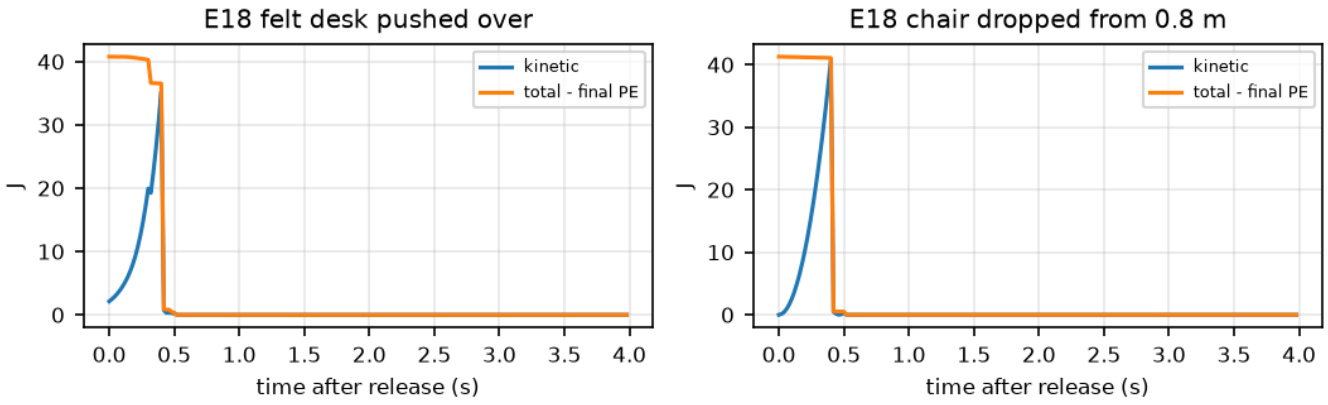


E16 spill

E16: each book's slip in the box against the desk tilt (dashed: atan 0.30).

E18: impacts and energy (assistance: phantom push / placement)

- Felt desk pushed over at its top edge by the frictionless phantom, then free: it fell onto its side (88°); mechanical energy never rose above its value when the push ended (largest gain 0.0001 J of 66.2 J) and it was at rest 0.53 s later. On the way down it struck desk_14 and shoved it 17 mm and 2.9° (bystander record).
- Chair released from 0.80 m: landed upright; no energy gain (0.000 J), at rest after 0.53 s.
- **The impact itself.** The checks above pass for any dissipative contact, so E18 also states what the contact model predicts and checks it: the contacts are critically damped (solref 4 ms, damping ratio 1), a restitution of about 0, **an unsourced modelling choice**; real chairs and desks bounce a little, and no restitution measurement for furniture on vinyl tile was found. Expected: ≥ 90 % of the impact's kinetic energy gone within 20 ms and no rebound of 5 mm or more. Measured: 98.3 % (desk, 36.1 J) and 98.9 % (chair, 41.2 J) dissipated, but the body's origin rises again by 5.0 mm (desk) and 8.2 mm (chair) after the impact, so the "no rebound" check fails: what follows the first contact is a small rock or bounce, not a perfectly plastic stop. For desk arranging (slow pushes, nothing dropped) this does not matter; a policy that knocks furniture over sees nearly plastic landings.



E18 energy

E19: legal-action abuse (assistance: none, public backend)

Random commands through the public backend (`registry.create`, no rendering) from the straighten-row start, 0.25, 0.5 or 1 s each: base twists anywhere up to 1 m/s and 1 rad/s, Spine targets anywhere in 0–0.85 m (those above the 827 mm stroke are refused), joint targets drawn anywhere within an arm's limits, Cartesian deltas up to ±0.15 m (many unreachable or self-colliding: refused before any physics) and gripper position or force commands. Two runs.

Seed	simulated	actions	refused	protective stops (joint torque / hand force / velocity)	furniture moved (largest)	peak object speed	fault
1	120.75 s	294	84	205 (110 / 92 / 3)	8 items (1.30 m)	0.60 m/s	none
2	120.25 s	239	37	122 (44 / 78 / 0)	23 items (2.33 m)	1.23 m/s	none

The robot shoves desks and chairs across the room; nothing exploded, every state stayed finite, no solver warning. Refusals include base commands during an arm's protective-stop hold (the interlock) besides unreachable IK targets and the Spine stroke; the three velocity stops of seed 1 came from random joint targets after collisions. The peak object speed is sampled at the end of each action (the 20 m/s fault guard checks every 20 ticks). Physics ran at 7.7× and 7.9× wall/sim, contact-heavy.

8. Robot demonstrations (assistance: scripted operator through the robot interface; E12's flip and E22 are phantoms)

Each video is rendered from the recorded states of one uninterrupted rollout, with a six-frame contact sheet; captions name the experiment, the assistance and the operator's step. The operator plans from ground-truth poses; every force on the furniture comes from robot contact unless marked phantom. All demonstrations run with the default `contact_x5` reflex profile.

Arm motion planning (`planner.py`, used by every demonstration): each arm moves alone; IK solutions are tried nearest first until one is clear of the robot and the scene (a scratch MuJoCo collision pass on the live state, 8 mm margin, contact allowed only with the object about to be touched), and the path is a direct joint segment if it is clear, else RRT-Connect with shortcut smoothing, streamed at ≤ 0.8 rad/s. Approach poses keep the hands about 0.9–1.0 m ahead of `base_link`: closer, every IK solution folds an arm into the Spine mount. Moving along straight lines instead makes the arms strike their own links and the Spine.

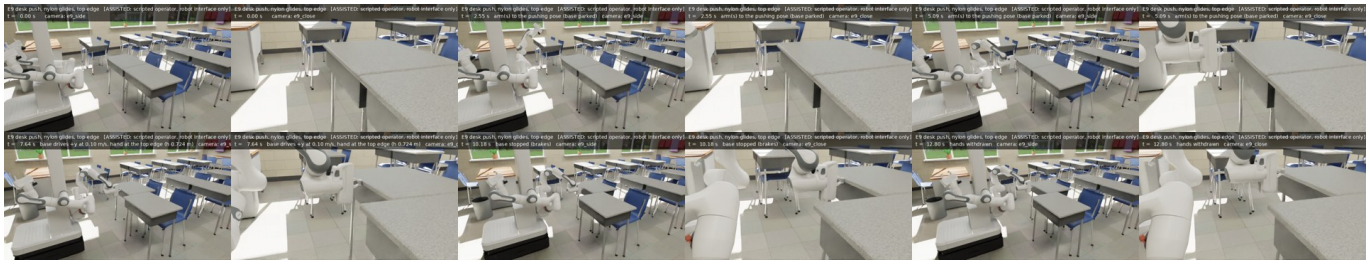
Demo	Task it supports	Result	Hand-force limit used (of 100 N)
E9 desk slid by one hand	straighten-row, exam-row	nylon desk 0.298 m at 20.9 N median (μ_k W 20.9 N); felt 30.7 N (31.0); felt pushed at the legs 31.8 N	49 N peak
E9 control, stock franka_ros thresholds	—	the 20 N hand-force reflex fires once and the interlock stops the base: the desk moves 2.5 mm	20 N threshold
E10 desk turned in place	straighten-row, make-pod	85.9°, tilt 0.0°	—
E11 one end lifted and dragged, three books inside	make-pod, contents variation	front glides 23.5 mm off the floor, tilt 3.4°, dragged 0.460 m on two glides; books shifted ≤ 1 mm in the box	— (joint torque 25 N·m, above franka_ros's 18–20 N·m)
E11 control, 8 N grip	—	**prediction failed:** the desk still lifts (front glides 5.2 mm clear, tilt 0.8°): the lower fingers carry the 25 mm edge from below	—
E13 chair pushed in	push-in-chairs	tucked (seat 0.415 m under the top, yaw 2.5°), 10.5 N median push (μ_k W 10.1 N)	—
E14 desk pushed into its pod neighbour	make-pod, group-work	edges in contact (gap −1.2 mm), hand force 18.2 N for one desk then 47.3 N for two (expected 20.9 / 41.7 N), desk-to-desk 28.4 N	36 N peak
E15 62.6 kg teacher's desk, rear bumper	move-teacher-desk	slid 0.646 m, yaw 0.3°, tilt 0.0°, at a mean 147.7 N while sliding (desk + the teacher's chair it pushes 0.30 m: μ_k ΣW 129.8 N; breakaway 144.3 N); no arm contact, no reflex, nothing else moved	— (arms not used)
E15 two hands (`teacher_hands`)	—	failed under the base interlock: 0.042 m in 16.5 s; at breakaway a hand's estimate passes 100 N 14 times and each time the base stops	99.6 / 98.7 N peaks
E17 chair pulled out and turned	seat-tutoring-group, rows 007/008	pulled 0.448 m (no longer tucked), turned 89.4°, tilt 0.0°, desk and every other item still (bystander check)	17 N peak
E12 chair lifted and carried by two arms; flip onto its desk by phantom	chairs-up	lifted 0.294 m, carried 0.834 m, set down upright; inverted on its desk, COM 86 mm inside the edge; knock: chair slipped 1.3 mm	—
E22 chair lifted off a desk and set down (phantom)	chairs-down	the desk moved 0.0 mm during the lift-off and in total; chair upright at rest behind the desk	phantom
E23 tote and wastebasket slid	clear-floor	tote 0.501 m at 24.5 N (μ W 26.1 N); wastebasket 0.459 m at 2.3 N (3.4 N)	—

Bystander check. Every demonstration records how far every item it does not mean to move has moved (> 10 mm or > 2° fails its acceptance, > 2 mm is listed), so a demonstration cannot pass while it shoves furniture it does not look at. No item moved or was nudged in E9–E11, E13–E15, E17, E22 and E23; in E12 the robot part moved nothing and the phantom knock moves desk_14 20 mm.

E9: sliding a desk with one hand

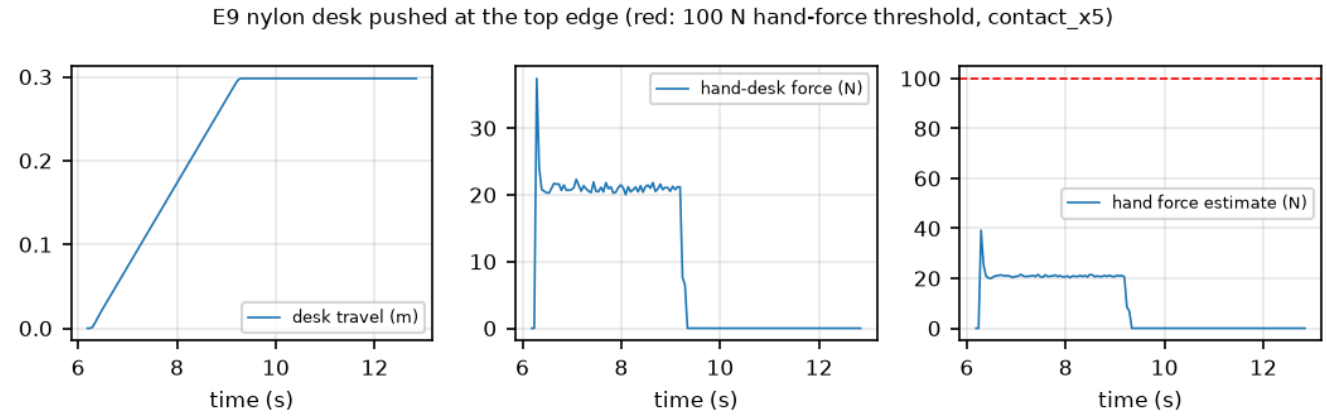
The base drives at 0.10 m/s with the right hand (fingers closed) against the middle of the desk top's front edge (h 0.724 m), or both hands against the front legs at 0.30 m (`felt_low`). Statics with a frictionless push predicts a felt desk tips at the top edge (μ_s 0.35 > d/h 0.301), but the rubber fingertip pads also press down through friction (F_v = μ_pad F_h), which raises the tip force to 62 N, above the 37 N breakaway, so the robot's push slides it, as observed. Push force (hand–desk contact) while sliding: nylon 20.9 N median (μ_k W 20.9 N), felt 30.7 N (31.0 N), felt at the legs 31.8 N; desks slid 0.30 m without tipping (≤ 0.01°). **Stock-threshold control** (`--reflex-profile franka_ros`): the 20 N hand-force reflex fires as the push approaches breakaway (25 N) and the base interlock stops the base: the desk moves 2.5 mm (without the interlock the base pushes on: 71 mm and 15 reflexes). With the

robot's stock collision behaviour a one-hand desk push is not possible, which is why the environment's default raises the thresholds (section 3.1).

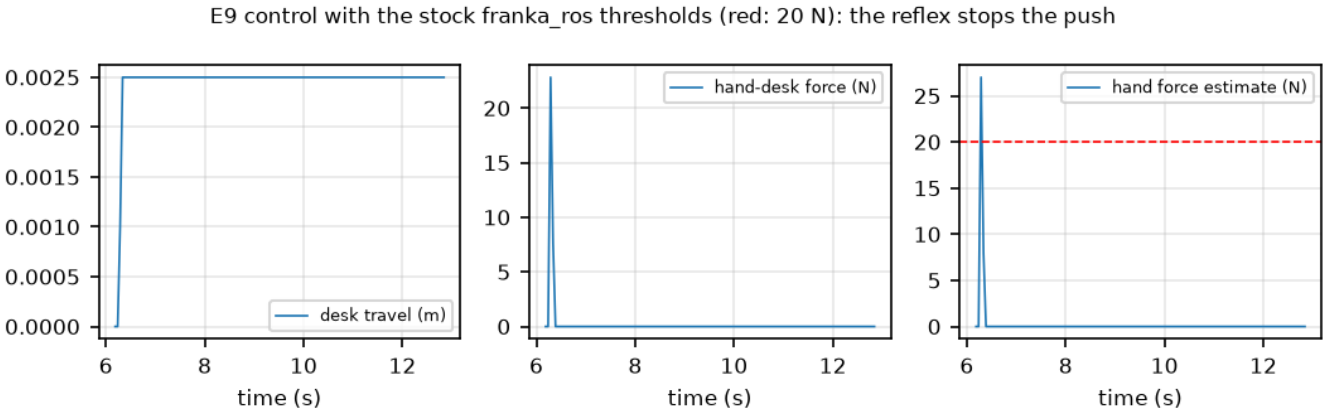


E9

E9, nylon desk pushed at the top edge: six frames in order.



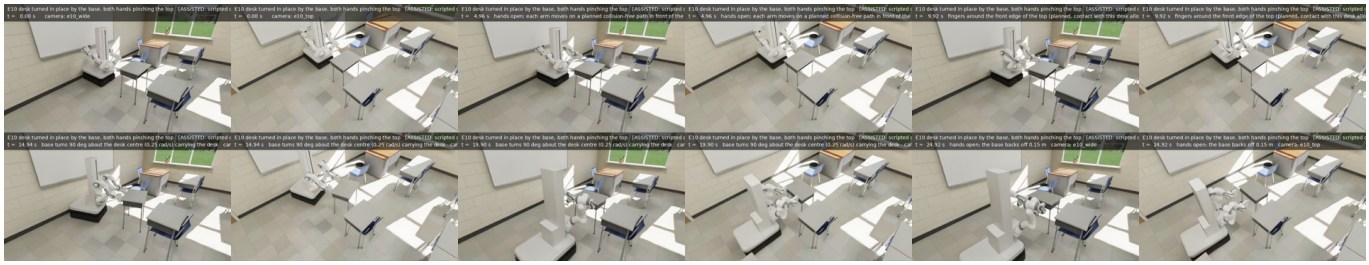
E9 timeline



E9 stock control

E10: turning a desk in place

Both hands pinch the front edge of the top (force mode, 60 N per hand) and the base drives a quarter circle about the desk's centre (0.25 rad/s): the desk turned **85.9°**, moved 40 mm, tilt 0.0°, no reflex; pad forces up to 86 N.

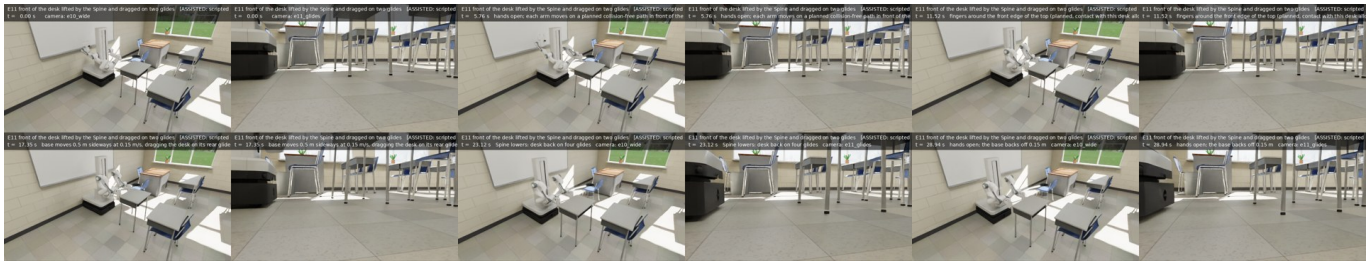


E10

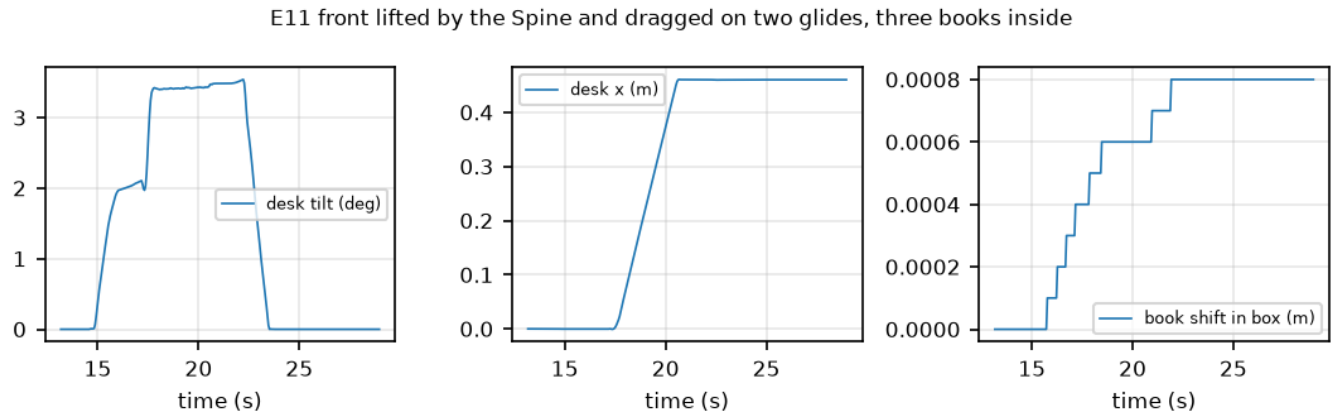
E11: lift one end and drag on two glides, with books

Three textbooks lie in desk_13's own book box. Both hands pinch the front edge; the Spine rises 40 mm in 3 s: the front glides leave the floor (23.5 mm clear, tilt 3.4°); the base drags the desk 0.460 m sideways on its rear glides

and sets it down (final tilt 0.0°). The books shifted at most 1 mm in the box (below the 16.7° friction angle, E16). Peak filtered joint torques 24–25 N·m: above the stock franka_ros thresholds, below the default's. The video's low view along the glides shows the drag; the 23 mm clearance is small even there. **Weak grip** (`--grip 8`, 8 N per hand; planned as a negative control): predicted not to lift; **it lifts** (front glides 4.9 mm clear, tilt 0.7° during the drag). With the arms holding their commanded pose, each hand's lower finger sits under the 25 mm top and carries the edge from below as a support, which needs no grip friction. The run shows that a vertical pinch lifts a desk end through support from below, even at 8 N (`weak_grip_still_lifts`, the prediction recorded as failed). No control that isolates grip friction (e.g. a horizontal pinch on a leg) was run.



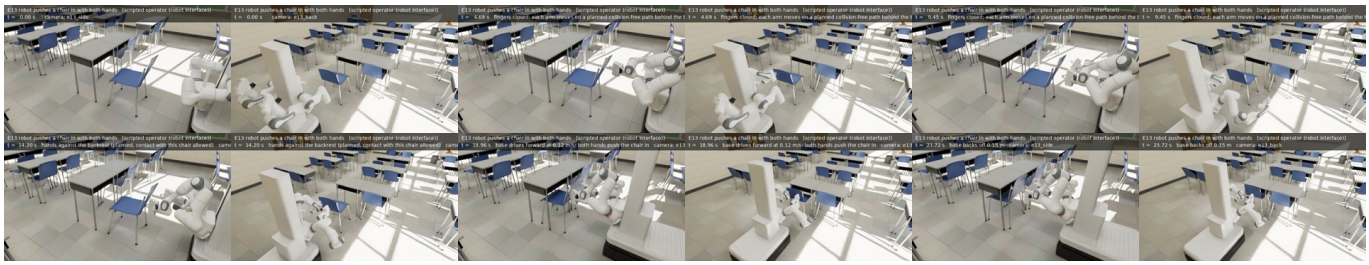
E11



E11 timeline

E13: pushing a chair in

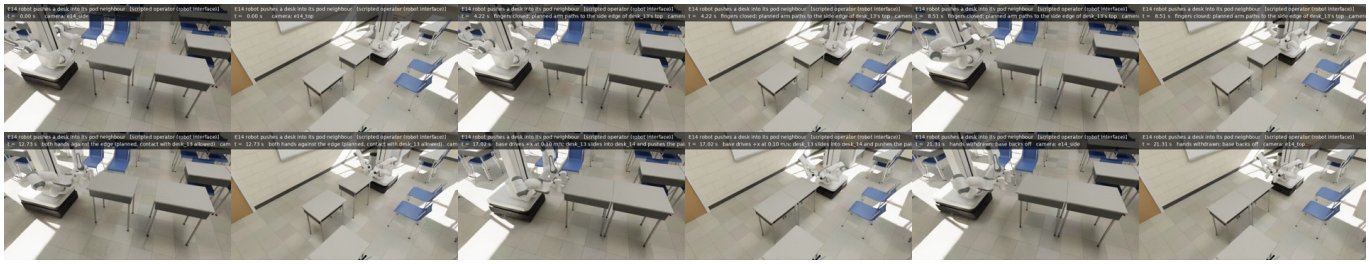
Both hands (fingers closed) stop 15 mm behind the backrest at 0.60 m and the base drives forward: the chair is pushed in until the evaluator scores it tucked (seat 0.415 m under the top, lateral −3 mm, yaw 2.5°), upright; nothing else moved. Median hand force 10.5 N against μ_k W = 10.1 N for the 5.26 kg chair. The hands stop short of the backrest because, with the arms holding their goals exactly, goals on the backrest press and turn the chair 7° before the push.



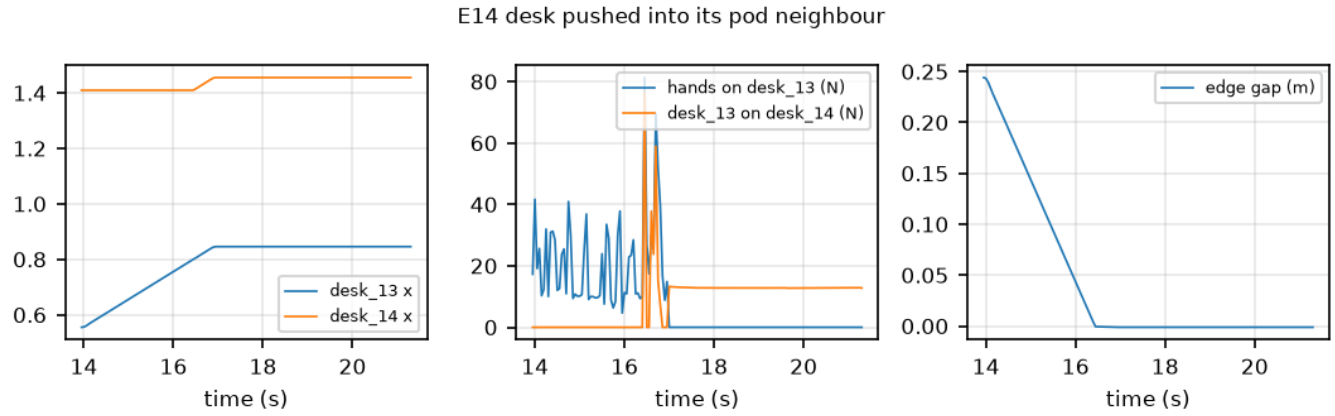
E13

E14: desk pushed into its pod neighbour

Pods need desk-to-desk contact (front edges ≤ 10 mm apart). desk_14 stands 0.25 m beside desk_13; both hands (fingers closed) push desk_13's side edge and the base drives 0.29 m: desk_13 slides into desk_14 (hand force 18.2 N, μ_k W 20.9 N), then pushes the pair 46 mm (47.3 N, 2 μ_k W = 41.7 N; desk on desk 28.4 N); final edge gap −1.2 mm (in contact, within the pod rule), both upright.



E14

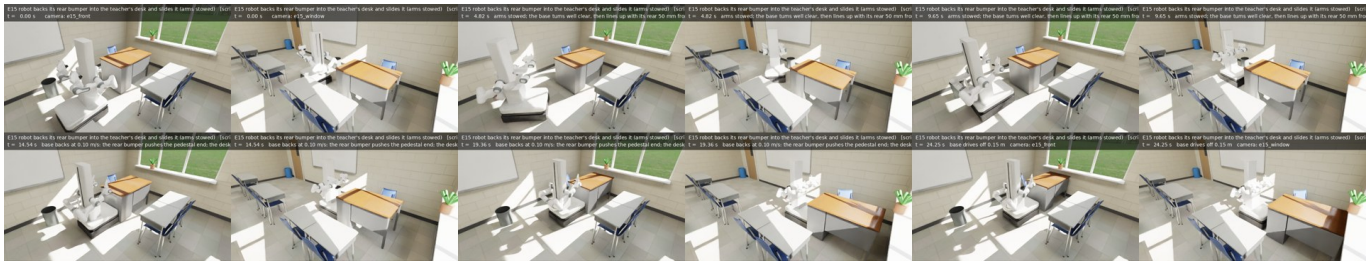


E14 timeline

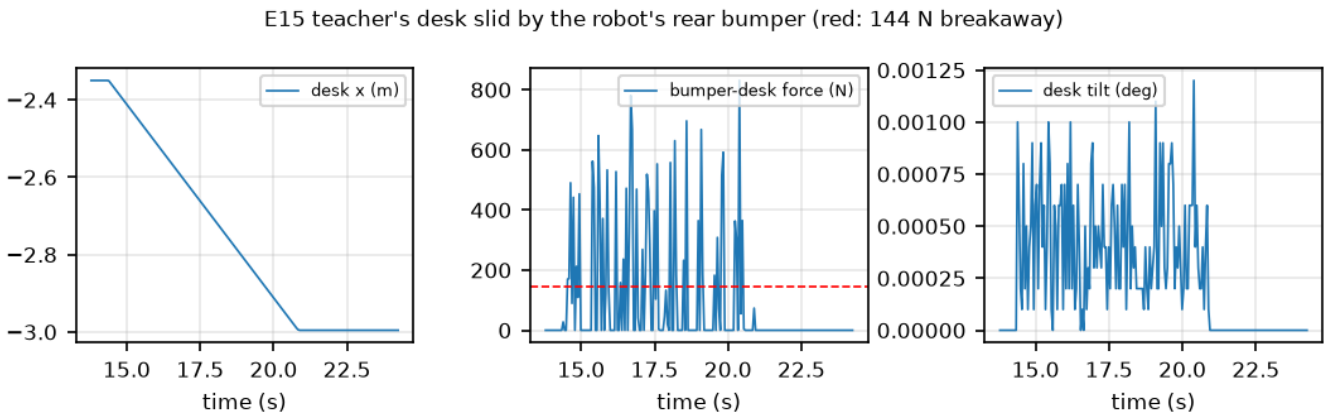
E15: sliding the 62.6 kg teacher's desk

The rear bumper. The robot turns well clear of the desk, lines its rear up 50 mm from the pedestal end and backs at 0.10 m/s with its arms stowed. The arms stand 0.16 m inside the rear bumper (0.24 vs 0.40 m behind `base_link`), so only the chassis touches; the two ArgoDrives push at up to 16 N·m each (E5's rear-bumper wall push). The desk breaks away and slides 0.646 m, pushing the teacher's chair behind it 0.30 m, square (yaw 0.3°), flat (tilt 0.0°), with no arm contact, no protective stop and nothing else moved. The bumper–pedestal contact is impulsive (a stiff hull on a box: single samples reach 849 N), its mean while sliding is 147.7 N against μ_k ($W_{desk} + W_{chair}$) = 129.8 N and the 144.3 N breakaway. Whether a real integrator lets the base push furniture depends on the nanoScan3 safety fields, which are not configured on delivery; the robot here does what its drives allow. This is the component the task needs (a 0.6 m slide); the task's target also turns the desk 90°, which was not demonstrated.

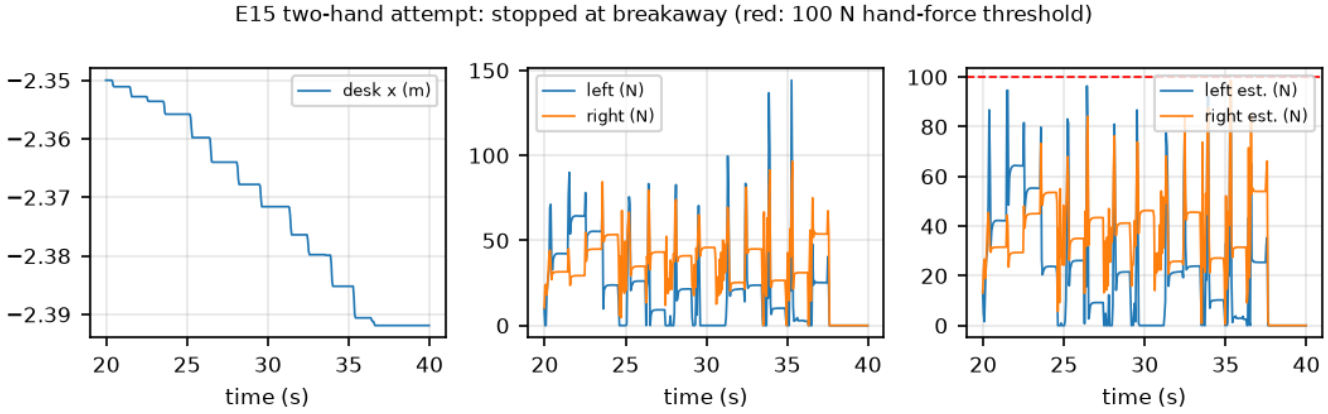
Two hands (`teacher_hands`). Both hands (fingers closed) against the pedestal end at 0.68 m, placed 15 mm short and then stepped 2 mm at a time until each touches; the base then drives at 0.04 m/s, yawing toward the less-loaded hand from the robot's own hand-wrench estimates. Under the base interlock the push fails: as the desk is about to break away the left hand's push-axis estimate passes the 100 N threshold 14 times (with 30–57 N of off-axis load on its other two axes), and each time the interlock stops and holds the base; in 16.5 s the desk moved 0.042 m. The contact forces of both hands reached the 144 N breakaway several times, so this is the operator's load sharing failing, not the world: a two-hand push within the default limits is not shown, not disproven. Without the interlock the arms, stopped by their reflexes, act as struts while the base drives on.



E15



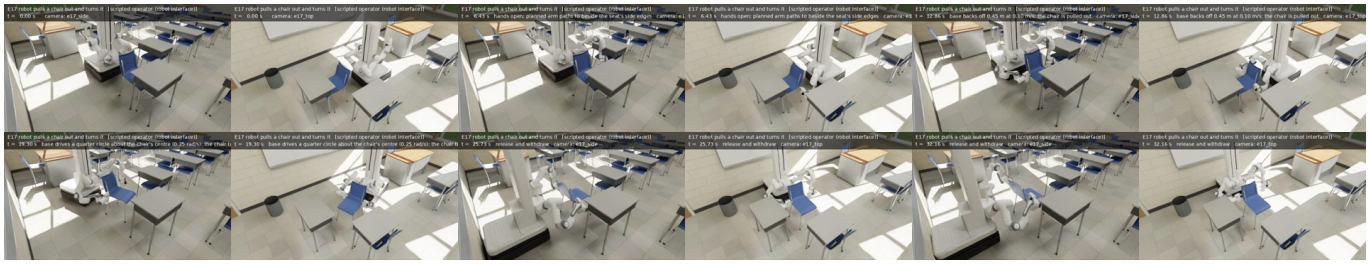
E15 timeline



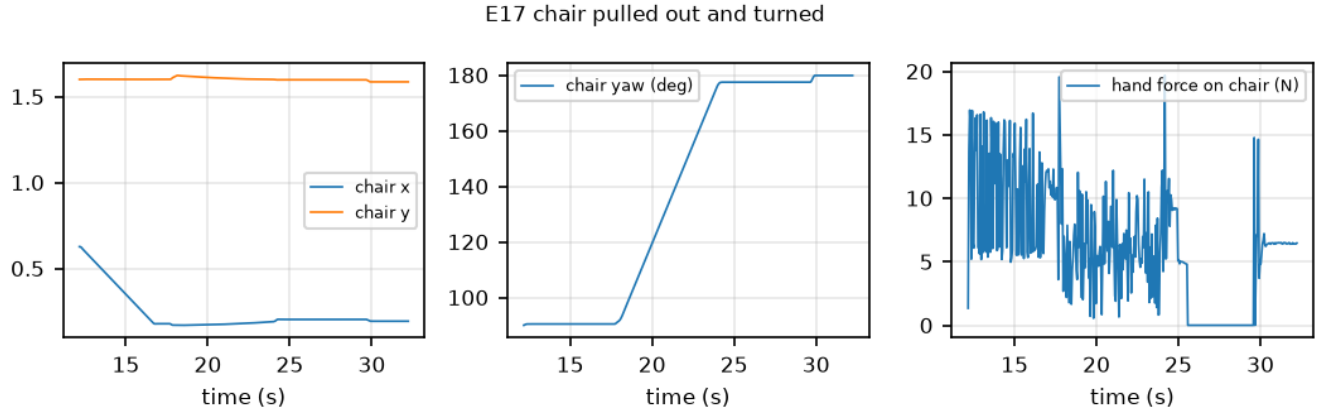
E15 two-hand attempt

E17: pulling a chair out and turning it

For seat-tutoring-group and action rows 007/008 ("pull a chair out"). The chair stands pushed in under desk_13 (seat front 0.12 m under the top, which the 0.30 m tuck criterion does not count as tucked; the pull mechanics do not depend on it); both hands pinch the seat's side edges in its rear half (force mode, 60 N per hand), the base backs off 0.45 m: the chair slides out (0.448 m, no longer tucked, tilt 0.0°), then the base drives a quarter circle about the chair's centre and turns it **89.4°**; release, hands up and away. Median horizontal hand force while pulling 11.1 N against μ_k W = 10.1 N. The scene stands ≥ 0.4 m clear of the teacher's desk, so the 0.45 m back-off cannot drive the robot's rear caster and rocker into it, and the bystander check shows that no item moved. Other grasps fail: pinching the backrest's top edge rocks the chair 16° (a pull at the top rail tips it, E4) and trips the hand-force reflex; pinching the seat's back edge wedges the hands under the backrest; and with the chair fully pushed in, the hands beside the seat stand between the desk's legs and drag the desk 0.40 m.



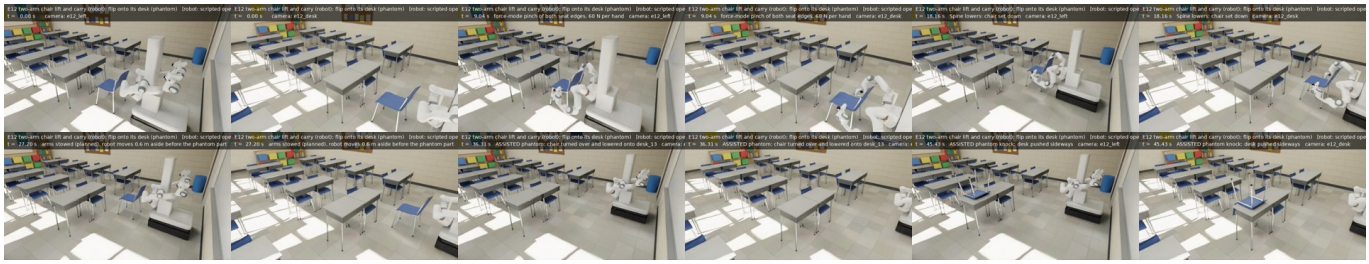
E17



E17 timeline

E12: two-arm chair lift and carry (robot), chairs-up placement (phantom)

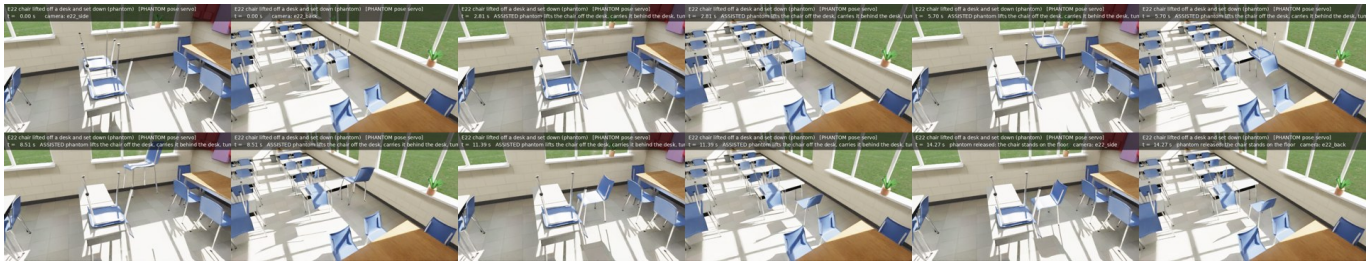
Both hands pinch the seat's side edges level with the centre of mass (fingers vertical, 60 N per hand); the Spine rises 0.30 m: the chair is lifted 0.294 m (each hand carries ~26 N, inside the FR3's 3 kg rating), carried 0.834 m sideways and set down upright. Then the **phantom** pose servo turns it over and lowers it onto its own desk_13: it rests inverted (180°), centre of mass 86 mm inside the top's edge, not touching the floor; an EN 1729-2-Annex-A-style knock (the desk pushed sideways at 1.5 m/s² for 0.2 s, phantom) moved the desk 30 mm (and with it desk_14, 20 mm) and the chair 1.3 mm relative to it. A robot regrasp-and-flip was not attempted.



E12

E22: chairs down: lift-off without moving the desk (phantom)

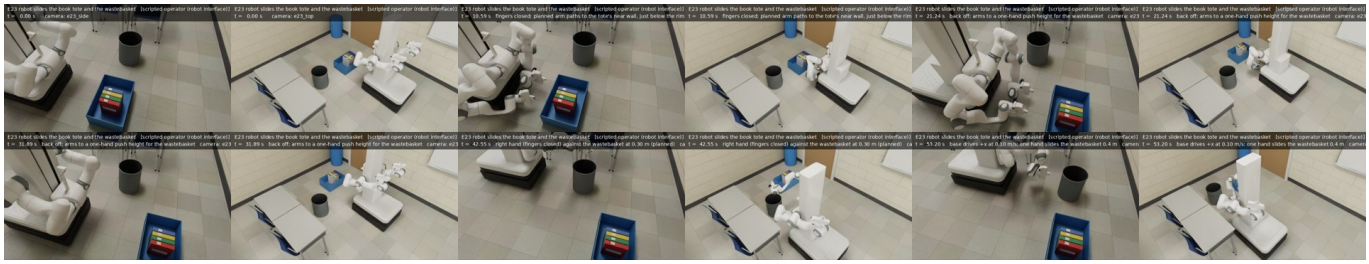
chair_41, upside down on desk_41 in the chairs-down start, is lifted 0.30 m by a phantom pose servo, carried 0.75 m behind the desk, turned upright about its centre of mass and set on the floor. The desk did not move (0.0 mm during the lift-off and in total, tilt 0.0°): the lift-off breaks only the seat-on-top contact. The chair stands upright at rest. (chair_13 is not used: this seed leaves it tucked under its desk, and lifting it catches the book box and drags the desk 0.41 m.)



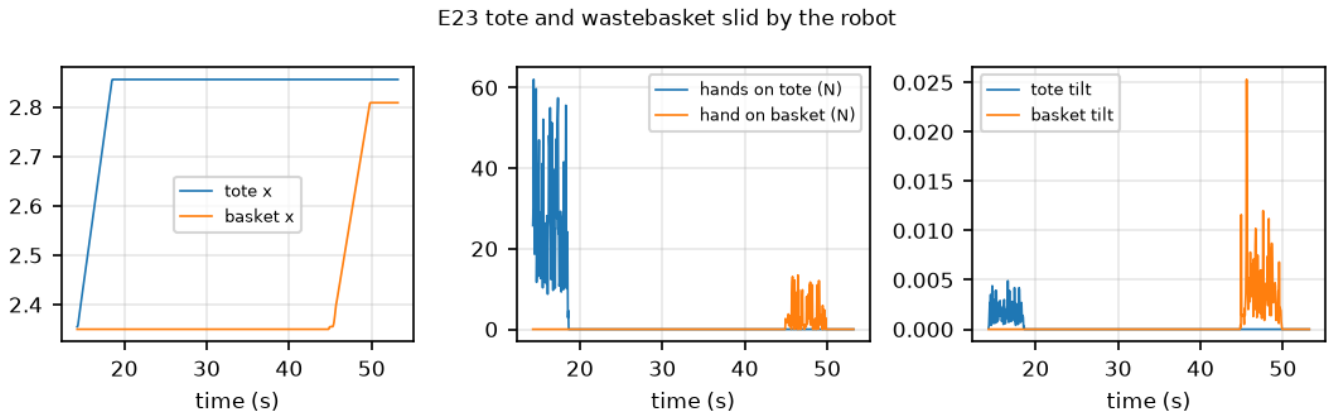
E22

E23: clearing the floor: tote and wastebasket

Both hands (fingers closed) against the 7.6 kg tote's near wall 20 mm below its rim (0.18 m: the lowest the arms reach about 0.9 m ahead of the base), the base drives 0.5 m: the tote slides 0.501 m at a median 24.5 N (μ W = 26.1 N, MU_TOTE 0.35), upright. The base then lines up 0.95 m short of the wastebasket and one hand pushes it at 0.30 m: 0.459 m at 2.3 N (3.4 N), upright. (At 0.64 m from the base the planner finds no collision-free way to reach 0.30 m height: the arm folds into the base.)



E23



E23 timeline

9. Tasks, instructions and the evaluator (`task.py`)

The evaluator reads only the world state; the policy never sees it (owner `GET /v1/evaluation`). Criteria per item, combined per task (all must hold; `score` = fraction met):

Every threshold below is the one `task.py` applies (values in `params.EVAL`):

- **upright** (for every checked item): tilt ≤ 3° (`upright_deg`) and every glide within 1 mm of the floor.
- **at rest**: the item's origin moved ≤ 2 mm (`rest_motion`) over the last 1.0 s of recorded history (`window_s`), and at least 1 s of history exists (so a desk coasting through its dots does not count). There is no separate angular criterion; a turning item moves its origin.
- **on_marks**: every glide within 30 mm of its own floor dot (`glide_tol`; matched to the desk's own glides, so a desk turned 180° onto swapped dots fails), yaw within 3° (`yaw_tol_deg`), upright, at rest.
- **tucked**: seat front ≥ 0.30 m under the desk top (`tuck_depth`; fully tucked is 0.418 m, the backrest 20 mm from the edge), lateral offset ≤ 0.08 m, yaw ≤ 10° from the desk's, upright. A looser 0.10 m would let a chair pulled 0.30 m back count as tucked while it stands 0.39 m into the 0.94 m walkway the instruction says to clear; E20 has a chair 0.25 m out, which must fail.
- **at_table** (tutoring): seat front 0.20–0.35 m from the table edge it faces (`ready_range`), facing it within 15°.
- **pod**: each facing pair's front edges −2 ... +10 mm apart (`pod_gap`; not overlapping beyond contact) with lateral offset ≤ 30 mm (`pod_lateral`); **exam**: every pair of the row's chair centres ≥ 1.25 m apart (`exam_pitch`: the UK JCQ's 1.25 m candidate spacing, 11.10 [S2]; no U.S. source found gives a seat pitch, so this is an authored spacing with a published precedent from another system, not a U.S. rule).
- **up_on**: chair inverted (tilt ≥ 175°, `inverted_deg`), its centre of mass ≥ 0.05 m inside its own desk's top (`inside_edge`), its lowest point no more than 0.40 m below the desk top, not touching the floor, and its backrest over the student side as the instruction says (`up_side_deg` 30°).
- **route**: 2 cm floor grid, a cell free when it is ≥ 0.4475 m (half the 0.915 m route, less half a cell) from every item footprint, wall, bookcase and cubby; a path from the door to the back must exist (clear-floor; group-work).
- **no_toppled_or_blocking**: nothing tipped more than 30° (unless being up on a desk is its target, or it started inverted) and nothing in the door swing, everywhere in the room.

E20 minimal pairs (17/17 correct): correct finished states pass for straighten-row, exam-row, make-pod, chairs-up and clear-floor; each of these fails: a chair pulled 0.25 m back from tucked; a chair up on its desk with the backrest over the board side; a desk turned 180° onto swapped dots; 45 mm off its dots; a chair only partly pushed in; a desk on its side over its dots; a correct placement read before a full 1 s rest window; an exam desk on its dots but turned 4°; middle exam chairs 0.62 m apart; a pod's back pair 30 mm away; a chair up on the wrong desk; three chairs across the three aisles. (A "desk on its spot but yawed 5°" case in a pair is pushed back to 2° by its neighbour in the settle, since a 10 mm gap cannot hold a 5° turn, so the exam-row case is used, where desks stand

alone.)

group-work. Instruction: "Group-work day: turn the two front rows (rows 1 and 2) into three pods of four, one for each pair of desks and the pair behind it. In each pod the pair nearer the board faces the pair behind it, and the four desks meet front edge to front edge on the pod's floor dots. Give each desk a chair on the outside of the pod, facing it and pulled out ready to sit. Leave a 0.915 m (36 in) wide route clear from the door to the back of the room, and leave the rest of the room as it is." Its evaluator scores places, not named furniture (`Evaluator.by_place`, `params.EVAL_GROUP_WORK`), because the 24 student desks, the chairs and the green dots are identical and unlabelled:

- **slot** (12): a pod place holds a desk, any of the 24, with every glide within 30 mm of the place's dot and yaw within 3° of the place's facing (the board-side pair turned toward the pair behind), upright, at rest. Which desks were turned does not matter, so a finished room built with any desks passes.
- **seat** (12): a chair, any of the room's identical chairs, at the desk in that place, on the outside of the pod, facing it within 15°, lateral ≤ 0.08 m, its seat front from 0.15 m under the top edge to 0.35 m beyond it (centre 0.294–0.794 m from the desk centre), with no other desk it stands behind and faces nearer, upright, at rest. This window includes chairs pulled out to the catalogue's own ready-to-sit distance.
- **pod_0..2**: in each column, the closest approach of the two facing front edges over the stretch both edges span is −2…+10 mm (contact anywhere counts); their widest separation there is at most 41.9 mm (the 10 mm closest-gap limit plus the desk width × sin 3°, the opening of one desk at the slot's full 3° against a square partner; a pair touching at one corner with that twist opens 31.9 mm); centres within 60 mm sideways (the two desks' 30 mm dot tolerances). A check on the centres' distance along y would fail two desks touching at one corner with a 2° twist that the slot check accepts. A 30 mm sideways limit would fail facing desks each 16 mm off their dots in opposite directions, and without the widest bound a V-gap 3.8 mm at one end and 57 mm at the other would pass.
- **rest_of_room**: every place the pods do not use (the desks and chairs of rows 3–4, the tutoring table and its chairs, the teacher's desk and chair) still holds one of its kind where one stood, upright: chairs within 0.10 m and 10°; desks, the teacher's desk and the table within 0.06 m and 5° (push-in-chairs' "do not move the desks" is 0.05 m / 5°; at 0.10 m / 10° 36.7 % of straighten-row's "drifted" desks and a desk pushed 0.095 m into an aisle would count as "as it is"; the route check only catches a route closed everywhere, not a narrowed aisle); the tote and wastebasket within 0.20 m, tilted ≤ 10°.
- **route**, **no_toppled_or_blocking** as above. The route's 2 cm grid with half a cell's allowance passes every aisle of 0.915 m or more and none below 0.895 m; between those the result depends on where the aisle falls on the grid. The pods keep the L0 aisle columns, 12–13 mm over 0.915 m, so desks inside their 30 mm dot tolerance can still close the route; both must hold.
- **Matching**: desks are assigned one-to-one to the 12 slots and the 12 places of rows 3–4, then chairs to the 12 seats (at the desks the slots received) and the 17 other chair places, each by the assignment with the most credit (`scipy.optimize.linear_sum_assignment`).
- **Score** (group-work only; every other task keeps the fraction of criteria met): 1 on success, otherwise 0.9 × progress × penalties, with progress = 0.60 × mean desk-place credit + 0.15 × mean credit of the six front-edge meetings + 0.25 × mean chair credit.
  - A desk's position credit rises linearly from 0, when its worst glide is 0.471 m from the place's dot (the move from its row; an untouched row desk, or a desk on the dots facing the wrong way), to 1 within 30 mm.
  - The six board-side places face the other way from every desk's start, so a desk there is worth 0.3 × a turn credit + 0.7 × the position credit. The turn credit is 1 when the desk faces the place's way within 3° and 0 when it is 90° or more off, linear between, wherever the desk stands (upright). So the six half-turns every finished room needs are visible before the desks are back on their dots; the back places and rows 3–4 carry no turn term, so the untouched room still scores 0. 0.3 of the 0.7 positional share is an approach credit, the turn credit × a linear fall-off of the centre distance from 1 at 30 mm to 0 at 2.5 m (the front band where desks are turned), so a turned desk carried back toward the nearest open board-side place gains credit over the whole way, not only its last 0.471 m.
  - A meeting's credit is the smaller of its two desks' position credits times a linear fall-off over 0.471 m of how far the edges are outside the pod bounds (closest gap −2…+10 mm, widest gap 41.9 mm, sideways 60 mm).
  - A chair's credit is the position credit of the desk it serves times a linear fall-off over 0.267 m (half the chair's depth) outside the seat window and over 75° beyond the 15° facing, and 0 unless that desk is the nearest the chair stands behind and faces (not a chair still pulled out at another desk in between). Turned desks still in the staging area therefore earn their chairs and meetings nothing.
  - Near credit reads the poses only; the criteria need the rest window.
  - Rest of the room: progress less 0.05 (one desk place's share, 0.60/12) per whole place not as found, never below 0; each place is charged from 0 at its tolerance rising linearly to 1 at 0.30 m or 45° (tote and wastebasket: 0.20 m to 0.50 m; knocked over, or a place left empty: 1). This is a deduction, so the same displacement costs the same at any stage and one displaced place costs less than one unfinished board-side desk.

- Penalties multiply what remains. Route: ×1 at 0.915 m falling linearly to ×0.5 at 0.815 m, for the widest route width that still connects the door to the back (measured by bisection with the same grid test, only when 0.915 m fails; 0.815 m is ADA 2010 403.5.1's exception, the narrowest an accessible route may be, for 24 in at most). ×0.5 for anything toppled or in the door swing.
- The untouched room scores 0, and constraints already met at the start earn nothing. Constructed cases: E27.

- **Start**: the rows 1–2 chairs stand where the students left them, pulled out 0.48–0.66 m from tucked (seat front 0.06–0.24 m beyond the desk edge), up to 0.03 m sideways and 10° turned, drawn per seed and conflict-checked: clear of the desk's rear legs (a draw from 0.20 m can put a turned chair's seat in a desk leg, which `layout.conflicts` does not test, and the settle tips it) and at most 3 mm wider than the desk each side, so the aisles keep at least 0.922 m. Each seed of the 1000 catalogue rows of group-work therefore has its own start state.

Instructions (`layout.TASKS[...]["instruction"]`), e.g. straighten-row: "The desks in row 3 (the third row from the board) have drifted off their places. Put each of the six desks back on its floor dots (the coloured dots under its four feet), squared to the board, and push its chair in under the desk. Leave the rest of the room as it is."

10. Interfaces: GAR CLI, HTTP and dashboard

- `gar list | inspect | tasks | doctor`: environment, robots (mobile-fr3-duo default, mobile-fr3-single), ten catalogue entries and five selectors, dependency checks including the locked robot assets, the pinned exterior HDR and oak textures (which the first render downloads) and the Isaac interpreter. The G1 is refused before any simulator starts.
- Backend `NewtonClassroomArrangementBackend` through `registry.create("classroom-arrangement", ...)` (aliases `arrange-classroom-desks`, `classroom-desks`): actions `base_twist` (a shared action; codec, help text and the policy allowlist cover it), `named_joint_target` (arms, fingers, Spine; base joints refused: they belong to the base controller), `gripper` (position or force), `cartesian_pose`/`cartesian_delta` (IK in the world frame, self-contact-free solutions only, bounded search, relative commands from the commanded pose), `keyboard` (the shared hand key map: W/S/A/D/R/F translation, I/K/J/L/U/O rotation, Q/E fingers, H force grip, plus UP/DOWN/LEFT/RIGHT driving the base and Z/X turning it; a released base key stops the base). **Every command lasts ≥ 0.02 s** (40 ticks, for every action type; section 11). Every command is planned and validated before any physics; a refused command raises `ActionRejectedError` and changes nothing (including a base command during an arm protective stop's 1 s hold). Policy observation: joints (positions, velocities), robot status (odometry, measured and commanded twist, per-arm reflex hold, filtered external joint torques and hand-wrench estimates, commanded grip force, refused-target count) and the selected policy cameras; no furniture pose, contact or evaluator. Owner: full state, evaluation, in-memory checkpoints (restore is a discontinuity), diagnostic renders, and:
  - a **per-command sub-step trace** (`begin_action_trace`/`end_action_trace`, `classroom_arrangement/trace.py`) in GAR's shared `PhysicsTraceStore` (the deposit_cans / wine_pouring pattern), so `GET /v1/history/{id}/trace`, `/trace/events` and `gar export --include-traces` read it: a sample every 0.5 ms tick with the joint state (furniture free joints included) and targets, and every 10 ms the body poses and velocities, base pose and commanded twist, filtered external torques, hand-wrench estimates and grip forces; protective stops as events; bounded at 256 MB of payload, which at one sample per tick holds roughly the last 30 s of simulated time (older samples are evicted and their traces report `evicted_samples`; the history itself stays complete) and up to ~400 MB on disk; `recording=False` disables it. Recording costs about 20 % of the physics time;
  - the **variation protocol**: `variation_contract()` (selectors preset, task, seed 0–9999, glides, contents 0–3, and the current episode) and `select_variation()`, so `/v1/variations`, `/select` and `/start` work on a live session; a different variation rebuilds the room (bodies, books and dots differ), restarts the render worker and refreshes the capabilities' task note; `seeded_reset` is advertised; an invalid reset option is refused with `UnsupportedActionError` before anything changes;
  - **backend options**: `reflex_stops_base` and `policy_samples` are real options, and unknown options are refused with `UnsupportedConfigurationError`;
  - the **geometry routes**: `get_geometry_index` lists every Newton shape and body (pencap's rigid catalogue: type, local pose, size, friction, mass, COM, inertia) plus each furniture item with its body, shapes and current pose and the task's targets, floor dots and checks; `get_geometry("desk_13")` returns the item's authored parts (`furniture.Part`: kind, size, pose, mass, material, role, friction) and current pose; `get_geometry("<shape id>")` and `get_geometry_asset("<sha256>")` return a shape and a mesh;
  - `robot_configuration()` returns the shared robot handles, setup id and fingerprint;
  - `device` other than `cpu`/unset is **refused** (physics is CPU MuJoCo); qualified `gar.robot-setup.v1` objects are accepted (section 3).
- **Packaging**: `classroom_arrangement` and its `benchmark.json`, `variations.json` and documents are in the package. Checked from an installed wheel, run with the source checkout off `sys.path`: the module loads from the wheel, all 17 task catalogues load, `registry.create` builds the room (19.8 s), an action runs, the state is finite and the evaluator answers.

- **Tests** cover discovery, catalogue, layouts, the codec, and the public backend (base twist moves the base, out-of-limit twist and base-joint targets refused, Spine target reached and the 827 mm stroke enforced, force grasp set, checkpoint round trip, policy view without privileged keys), the infrastructure (setup objects, device refusal, G1 reason, robot configuration, geometry, the sub-step trace read back from GAR's shared trace store, the 0.02 s minimum for every action type, the variation contract and selection, bounded planning time) and a **single-arm** session (`mobile-fr3-single`: one arm, left-arm commands refused, right-hand Cartesian move, grip, base drive, finite state, evaluation).
- **HTTP and dashboard**, with the GPU shared: real `gar run classroom-arrangement --policy-key --port 0` sessions, keys read from the launcher's JSON in memory only, on **group-work** with `head_rgb` (a route-check task) and on **make-pod** with all three arm cameras (`--backend-options-json`, minimum command 0.06 s). Dashboard page, script and the calls it makes: 200. Policy: capabilities, task, observation and the `head_rgb` PNG 200; 0.5 s actions at 8.6–11.6× (10.4–16.0× with three cameras) wall/sim including the observation render; the shortest legal command of every type at 20.4–77.0× (17.1–35.0× with three cameras); a 1-tick command is refused (400); a batch of twenty minimum-length twists ran at 20.5× (14.2×); a Cartesian target 3 m away is refused in 0.08 s; checkpoint save and state 403. Owner: save, act, restore (the tick returns), state and evaluation 200; `GET /v1/history/{id}/trace` and `/trace/events` 200 with the sample count; `gar pause` then `gar export --include-traces` writes a complete export with no trace gaps; `GET /v1/variations` reports the selectors and the current episode, `select {seed: 4}` returns the reset request and `/start` rebuilds the room. All 16 checks pass in both sessions. The dashboard, connected once per principal in headless Chromium, shows the head camera, task and controls. `session.json` writes non-finite joint limits as `null`, the evaluation carries `success` for GAR's evaluation normaliser, and the render worker's idle timer is cancelled while an action runs.
- **Keyboard page** (`python -m classroom_arrangement.manual --session <id>`): on a live session, ArrowUp held for 1.5 s moved the base 0.39 m and R moved the right hand; physics advanced only while keys were held.

11. Runtime (E21)

Public path (`registry.create`, one RTX observation of the selected cameras after each action, 16 spp), measured **with the GPU shared** with another environment's render workers and liquid jobs (99–100 % utilisation for most of the day). These numbers are upper bounds for this machine; on an idle GPU a 640 × 360 frame takes 0.16–0.17 s.

0.5 s actions alternating base twists and Spine moves:

Cameras	create (build, settle, worker start, first frame)	wall/sim, first	warm mean	warm max	physics alone	render per action
head_rgb	47.0 s	7.91×	7.88×	8.4×	6.9×	0.49 s
head + both wrists	40.2 s	9.67×	8.93×	10.34×	6.91×	1.01 s

Physics alone runs at 6.5–8.0× wall/sim (the per-tick sub-step trace adds about 20 %, the arm compensation and feed-forward a few %, the two LIDAR scans 1.8 ms per observation). **The minimum command duration scales with the render load**: 0.02 s per 640 × 360 frame of rendered pixels per observation, so 0.02 s for `head_rgb` (or no camera), 0.06 s for three 640 × 360 cameras and 0.08 s for one 1280 × 720 camera (`capabilities.notes.min_command_s`; shorter commands are refused before any physics). A fixed 0.02 s floor would let commands reach 205× with three cameras or at 1280 × 720. Every action type at its session's minimum, through the public path:

Configuration	minimum	plain commands at the minimum	5 cm Cartesian deltas at the minimum (IK ≤ 1.4 s)	worst legal command	frame per observation
head_rgb 640 × 360, make-pod	0.02 s	15.9–32.1×	20.8–92.3×	92.3×	0.17–0.61 s
head_rgb 640 × 360, group-work (route evaluator)	0.02 s	19.3–28.1×	29.5–81.5×	81.5×	0.24–0.68 s
head + both wrists 640 × 360	0.06 s	15.9–38.1×	30.9–41.5×	41.5×	0.54–1.87 s
head_rgb 1280 × 720 (native)	0.08 s	17.2–23.4×	17.6–36.3×	36.3×	0.81–1.44 s

Every legal command in every measured configuration is within the 200× limit. Refusals cost ≤ 1.4 s (an orientation-unreachable delta from the stowed pose; a 2–3.6 m target is refused in 1 ms by the reach check, a 1-tick command in < 1 ms). This rests on: the evaluation is computed once per (tick, episode) and the route check uses connected components (0.011–0.015 s instead of 0.12–0.13 s three times per HTTP command); the IK search is budgeted at 0.8 s; the render worker's idle release is 300 s (after a longer pause the next image pays a ~6 s cold start and a ~1.8 s first frame). Through HTTP see section 10 (group-work with `head_rgb`, make-pod with three cameras). The four base cameras cost the same per frame as the others and count in the same render load.

12. Limitations and open issues

- **The teacher's desk is moved with the rear bumper, not the hands** (E15). The two-hand push stops at breakaway under the 100 N hand-force reflex and the base interlock (the operator's load sharing, not shown impossible); the rear-bumper slide works within every protective limit but relies on the base being allowed to push furniture, which on a real TMR depends on how the integrator configures the nanoScan3 fields. The task's 90° turn of the desk was not demonstrated.
- **Glide friction is the largest uncertainty.** The sled coefficients [F1] and a furniture-scale pull [F2] differ by about 2×; at the furniture-scale value a desk pushed at its top edge tips instead of sliding (E24). No physical calibration was possible. The static→sliding schedule (v_c = 2 mm/s) is a smooth stand-in for real stick-slip; for steel glides the stick-slip drops are 3–4× the friction step, and near the tip/slide boundary a large $\mu_s - \mu_k$ gap lets pivot glides creep (E24's three 6–9 % misses).
- **Sliding contact is intermittent** (section 6): a sliding glide carries no load in 20–61 % of the ticks at 50 mm/s and its trailing pair in 84–97 % at E4's speeds, at every step and contact time constant tried. Means are right; instantaneous contact forces are impulsive. In a fast push the trailing load goes inside the rigid-body band (felt, steel; nylon 7–9 % below it at the default step, inside it at 0.25 ms), but **once the desk rotates it reaches 1° at 0.61–0.81 of a rigid body's estimate** whatever the step (0.125–0.5 ms) or contact time constant (2–8 ms): cause not established; statics exact.
- **Glide footprints are authored** (whiteboard furniture; Virco's Revit families were not converted). E24 shows a 2 cm change moves one or two near-boundary outcomes.
- **Masses** follow one rule (Ezell's listed weight, else SchoolOutlet's); listed weights may include packaging, and the desk is the lowest of three listings (E24's −15 % and +26 % change no outcome; push forces scale with it). The Duo totals 216.8 kg against the manual's 225 kg (the URDF's TMR and Spine inertials were raised to the manual's subsystem masses with authored blocks; the remainder is unassigned).
- **Impacts are nearly plastic** (E18): critically damped contacts dissipate 98–99 % of an impact's energy within 20 ms (an unsourced modelling choice; no furniture-on-vinyl restitution source was found), and a 5.0–8.2 mm rise after the impact fails E18's "no rebound" expectation.
- **Coefficients not varied:** `MU_PLASTIC` 0.30 (desk on desk, E14), `MU_STEEL` 0.35, `MU_ROBOT` 0.40 (robot covers on furniture), `MU_WHEEL` 0.60 and the 10 ms reflex filter are labelled estimates with stated ranges; E24 does not vary them because none of its cases involves them, and no robot demonstration was repeated with them changed.
- **Robot limits.** The default reflex profile (`contact_x5`, franka_ros × 5) is an integrator's choice with a published precedent [D9], not a Franka default; with the stock thresholds (selectable) one-hand desk pushes stop at 20 N (E9 control). The FR3's real collision detection uses its own model-based estimates; here they are computed from contact forces and filtered. The arm compensation uses MuJoCo's bias force from the previous 0.5 ms step (exact model, one tick old), not an identified FR3 model. The stowed arms stand ~0.35 m out either side and the arm mount ~0.2 m ahead of the chassis, so sideways contact loads the arms first (E5) and the robot cannot push furniture with its chassis front (its rear bumper can, E15). The base sensors (section 3, E26) are observations only: the D455s give RGB, not depth; the LIDARs have no noise model and no safety fields; the IMU has no bias or noise. The drives' spring-applied holding brake (35.5 N·m) is not modelled; a parked wheel holds up to the motor's 28.4 N·m. Odometry is off by 3.8 cm after E5's free moves (0.47 m once the wheels have slipped against the wall). Casters are passive swivels without flutter.
- **Only this robot family.** Two qualified configurations (Duo, single arm) from the shared catalogue; `base.py` and `scene.py` name the TMR's joints and links (drive positions, caster/rocker links, wrist cameras on link7), so another mobile manipulator needs code, not just a catalogue entry. The arms are a copy inside the Duo URDF with libfranka's joint impedance, not the shared `fr3` model composed onto a base.
- **Humanoid: not built or tried.** The repository pins a velocity-commanded G1 walking policy (`football/newton15/balance.py`) and barcode_scanning stands a floating G1; a classroom G1 would need the room built around a floating humanoid, that policy behind `base_twist`, and evidence of standing, walking the aisles and pushing. None of it was attempted; the catalogue refuses `g1` saying so. Whether a G1 can do these tasks here is unknown, not disproven.
- **Operator, not policy.** The scripted operator plans from ground-truth poses with a collision-aware planner and hand-picked approach poses (hands ~0.9–1.0 m ahead of the base). Several demonstrations needed more than one grasp design (E17 lists three that fail). Nothing about it is available to a policy.
- **Phantoms where the robot was not shown:** the chairs-up flip onto a desk (E12) and the chairs-down lift-off (E22) are phantom pose servos; a robot regrasp-and-flip and a robot lift-off were not attempted. E16's tilt is a phantom.
- **E11 has no grip-friction control:** with the arms holding their pose the lower fingers carry the desk edge from below, so an 8 N grip lifts the desk too; no horizontal-pinch control was run.
- **Simplified bodies:** the tote and its four books are one rigid body (the books cannot shift or spill from it); the chair's collision shell is three flat panels and a seat box (the render draws a moulded shell); the teacher's desk

pedestal is one box (drawers render only); the wastebasket is a closed cylinder (the render shows it open).

- **Pull forces not separated:** E13's push and E17's pull are read from the hands' contact forces, which include each grasp's own load on the chair; the glide loads were not recorded.
- **Runtime caveats** (section 11): after `idle_release_s` (300 s) without a request the next image pays a ~6 s cold start; the first render of a new install downloads the pinned HDR and textures (needs network).
- **Visuals:** chair shells are a moulded approximation, not the 9018's exact shape; no students or wall-mounted TV; the outside is a dome HDR (a sports field), not geometry; the window glass is physical but not drawn (open panes); textbooks carry a plain title band, not cover art; sunlit shells glare.
- **Not built**: folding/pasteboard tables, chair stacking, the second (orientation-evening / assembly rows) preset, the Robotiq 2F-85 gripper option.

13. Running the environment

The experiments of sections 7, 8 and 11 were run with probe scripts that are not part of the release. The released environment is inspected, run and checked with the GAR CLI:

```
gar inspect classroom-arrangement                # robots, cameras, description
gar tasks inspect classroom-arrangement          # the ten tasks and the selectors
gar doctor classroom-arrangement                 # dependencies, robot assets, Isaac interpreter
gar run classroom-arrangement --task make-pod    # a session (--robot mobile-fr3-single for one arm)
python -m classroom_arrangement.params           # the parameter ledger of section 5
python -m classroom_arrangement.manual --session <id>   # keyboard page on a live session
```

14. Sources

[F] = fetched and read; [S] = only a web-search result or summary was seen (the value quoted is what that summary showed). Every key cited in this report or in `params.py` is listed; nothing else is.

Standards and practice

- [S1] EN 1729-2:2023, Furniture — Chairs and tables for educational institutions — Part 2: Safety requirements and test methods, Annex A (the E12 knock is "Annex-A style", not the standard's test). https://cdn.standards.iteh.ai/samples/69943/fd72a3d2f8f744359283a53d3c7f76cb/SIST-EN-1729-2-2023.pdf [S]
- [S2] JCQ, Instructions for Conducting Examinations, §11.10 (1.25 m between candidates). https://www.jcq.org.uk/knowledge-hub/intructions-for-conducting-examinations/ [S]
- [S3] 19 Tex. Admin. Code §61.1040 (general classrooms ≥ 700 ft²). https://tea.texas.gov/laws-and-rules/commissioner-rules-tac/coe-tac-currently-effect/ch061cc.pdf [S]
- [A1] U.S. Access Board, 2010 ADA Standards for Accessible Design, §403.5.1 (clear width of walking surfaces 36 in / 915 mm). https://www.access-board.gov/ada/ (literature value; the section text was not fetched) [S]

Furniture

- [V1] Virco, 785 Series Hard Plastic Desk with Metal Book Box. https://virco.com/product/785-series-hard-plastic-desk-metal-book-box/ [F]
- [V2] Ezell Company, 785MBBM (24 lb). https://ezellcompany.com/product/vircos-open-front-student-desk-with-solid-plastic-top-metal-bookbox-785mbbm/ [S]
- [V3] Schools In, 785MBBM (28.5 lb; 1⅛ in glides). https://www.schoolsin.com/vir-785mbbm.html [S]
- [V4] Furniture Tycoon, 785MBBM (30.20 lb). https://www.furnituretycoon.com/open-front-desk-with-metal-bookbox-hard-top-785mbbm/ [S]
- [V5] Virco, 9000 Series 4-Leg Stack Chair (polyethylene shell). https://virco.com/product/9000-series-4-leg-stack-chair/ [F]
- [V6] SchoolOutlet 9018 (10.417 lb); VastMarket VTCV-18 (11 lb item, 13 lb shipping; seat 15 × 16¼ in). https://www.schooloutlet.com/products/virco-9018-school-chair-for-classrooms-5th-grade-to-university-18-student-seat-height ; https://www.vastmarket.com/virco-vtcv-18 [S]
- [V6b] Ezell Company, Classic Series 18" Stacking Classroom Chair, by Virco #9018 (11.6 lb; 18¾ × 21½ × 30⅝ in). https://ezellcompany.com/product/vircos-classic-series-18-stacking-classroom-chair-9018/ [S]
- [V7] Virco, 4000 Series Table 30" x 60" Rectangle (1⅛ in particleboard-core laminate top). https://virco.com/product/4000-series-table-30-x-60-rectangle/ [S]
- [V7b] Ezell Company, 4000 Series 30" x 60" Rectangular Adjustable Activity Table, by Virco #483060 (71.3 lb). https://ezellcompany.com/product/vircos-4000-series-rectangular-30-x-60-adjustable-activity-table-483060/ [S]
- [V8] Virco, 540 Series Teacher Desk Single Pedestal (48 × 30 in; center, file and box drawers). https://virco.com/product/540-series-teacher-desk-48-x-30-laminate-top-single-pedestal/ [S]
- [V8b] SchoolOutlet, Virco 543 - 540 Series Teacher's Desk Single Pedestal, 30"D x 48"L Laminate Top (138 lbs, one carton; 1.125 in top; powder-coated steel). https://www.schooloutlet.com/products/virco-543-540-series-teachers-desk-single-pedestal-30-x-48-top [S]
- [V9] Virco Revit families (785, 9000 series; geometry not converted). https://virco.com/wp-content/uploads/785_desks.zip ; https://virco.com/wp-content/uploads/virco_9000_series--2.zip [F]

Friction

- [F1] Substratum Group (2018), Chair Glides: Using Physics to Evaluate Design and Risk (GS-1 sled SCOF/DCOF chart; the source of `GLIDE_MU`). https://www.substratumgroup.com/wp-content/uploads/2018/09/ChairGlideResearch2018.pdf [F]
- [F2] Yeow, MacRae, Dhakal, Bradley (2018), Validating the Sliding Mechanics of Office-Type Furniture Using Shake-Table Experiments, Bull. NZ Soc. Earthquake Eng. 51(1) (furniture-scale pull, μ_s 0.45 for a metal-base desk on vinyl; the upper end of E24). https://bulletin.nzsee.org.nz/index.php/bnzsee/article/view/41/27 [F]

Robot

- [D1] Franka Robotics, Prototype Manual and Datasheets: Mobile FR3 Duo (R04245: 225 kg, TMR 130 kg, Spine 45 kg, 827 mm stroke, 0.3 m/s). https://franka.de/hubfs/Prototype%20Manual%20and%20Datasheets%20_Mobile%20FR3%20Duo_R04245_EN.pdf [F]
- [D3] frankarobotics/franka_description, commit 7aeeddc449edf8d62b594f9e36a81da53e7796f9 (Apache-2.0). https://github.com/frankarobotics/franka_description [F]

- [D4] Neobotix/ebm-papst, ArgoDrive data sheet (AD-100.12: 16.0 N·m maximum acceleration torque, 28.4 N·m maximum braking torque, 0.8 m/s² at full load, 180°/s steering). https://www.neobotix-roboter.de/fileadmin/images/downloads/Datenbl%C3%A4tter/Data-Sheet_ArgoDrive_DE.pdf [S] (values checked against the datasheet table)
- [D6] Franka Robotics, Datasheet Franka Research 3 (R02212 v2.5, June 2026). https://franka.de/hubfs/Datasheet%20Franka%20Research%203_R02212_2.5_EN_June%202026.pdf [F]
- [D7] Franka Robotics, Franka Hand product manual 1.2 (30–70 N continuous, 80 mm stroke); Panda datasheet (140 N maximum, 50 mm/s per finger). https://download.franka.de/documents/220010_Product%20Manual_Franka%20Hand_1.2_EN.pdf ; https://www.generationrobots.com/media/panda-franka-emika-datasheet.pdf [S]
- [D9] boschresearch/franka_utils, `set_collision_behaviour` (scales the default force/torque thresholds by a ratio; README example 5×). https://github.com/boschresearch/franka_utils [S]
- [D10] Franka Robotics, franka_control documentation and `franka_combined_control_node.yaml` collision_config (upper torque thresholds [20, 20, 18, 18, 16, 14, 12] N·m; upper force thresholds [20, 20, 20, 25, 25, 25] N / N·m). https://frankarobotics.github.io/docs/doc/franka_ros/franka_control/doc/index.html ; https://github.com/frankaemika/franka_ros/blob/develop/franka_control/config/franka_combined_control_node.yaml [S]
- [D11] libfranka `examples/examples_common.cpp` `setDefaultBehavior` (collision thresholds 10/20 N·m and N; joint impedance 3000…2000 N·m/rad), as ported in `barcode_scanning` (repo). https://github.com/frankarobotics/libfranka [repo]

Software and assets

- [M1] MuJoCo documentation, Computation (contact model, friction priority, no-slip). https://mujoco.readthedocs.io/en/stable/computation/ [S]
- [H1] Poly Haven, Suburban Football Field HDRI (Grzegorz Wronkowski, CC0), 2k HDR, SHA-256 38d31700e0d22b36ac3b74ee15c67cdc2c793e3a88bcafa7c69b67abf58e743f. https://dl.polyhaven.org/file/ph-assets/HDRIs/hdr/2k/suburban_football_field_2k.hdr [F]
- Repository: the barcode_scanning and filleting environments (render worker, reflex, force-mode grasp), the whiteboard classroom (room shell, oak veneer), pencap's rigid geometry catalogue, the deposit_cans trace pattern.